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A novel RAB5 binding site in human VPS34-CII that is likely the primordial site in eukaryotic evolution

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eLife Assessment

This **convincing** study examines a novel interaction of RAB5 with VPS34 complex II. Structural data are combined with site-directed mutagenesis, sequence analysis, biochemistry, yeast mutant analysis, and prior data on RAB1-VPS34 and RAB5-VPS34 interactions to provide a new perspective on how RAB GTPases recruit related but distinct VPS34 complexes to different organelles. The judgment is that this work represents a **fundamental** advance in our understanding of VPS34 localization and regulation.

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Abstract

RAB5-GTP activation of the multiprotein VPS34 complex II (VPS34-CII) is critical for endosomal sorting and maturation, phagocytosis, and receptor downregulation. RAB5-GTP activates VPS34-CII, by binding to a helical insertion in the C2 domain of VPS34 on the BECLIN1/UVRAG-containing adaptor arm of the complex. The autophagy complex, VPS34 complex I (VPS34-CI), features a unique ATG14L subunit in place of the VPS34-CII UVRAG subunit, and we found that this distorts the adaptor arm to alter the VPS34 RAB-GTPase binding pocket so that it preferentially binds RAB1-GTP. Surprisingly, our higher-resolution single-particle cryo-EM structure of VPS34-CII showed a second RAB5-GTP binding site on the VPS15 solenoid region. This site (VPS15-RAB5-site) appears to be the primordial RAB5-binding region. A mutant in the helical insertion of the C2 domain of human VPS34 that mimics the *Saccharomyces cerevisiae* sequence abolishes RAB5 binding to VPS34. Mutation of the VPS15-RAB5-site ortholog in *S. cerevisiae* VPS15 resulted in defective CPY sorting, loss of colocalization with the RAB5 ortholog Vps21, and loss of binding to Vps21 *in vitro*. Evolutionary expansion from one to two RAB5-orthologue binding sites may have increased membrane binding and VPS34-CII activity to adapt to more complex endocytic systems.

Introduction

Mammalian cells evolved three classes of phosphoinositide 3-kinases (PI3Ks). The class I and II enzymes are important elements of signal transduction pathways that evolved in metazoa (Burke, 2018 ; Koch *et al.*, 2021 ; Kucukdisli *et al.*, 2023 ; Lo *et al.*, 2023 ; Lo *et al.*, 2022 ; Rathinaswamy & Burke, 2020 ; Vadas *et al.*, 2011 ; Vogt *et al.*, 2023 ), but the class III PI3K, VPS34, is the primordial PI3K, which is present in all clades of eukaryotes. VPS34 catalyzes phosphate transfer from ATP to the D3-OH of phosphatidylinositol, producing phosphatidylinositol 3-phosphate (PI3P), which is important for cellular sorting pathways, including maturation of endosomes, phagocytosis, and autophagy. Also, recent studies suggest that VPS34/PI3P-dependent pathways can be hijacked by viruses (Dahmane *et al.*, 2022 ; Gong *et al.*, 2024 ; Wang *et al.*, 2021 ; Williams *et al.*, 2021 ; Wu *et al.*, 2024 ). VPS34 is a component in two complexes, VPS34 complex I (VPS34-CI) and VPS34-complex II (VPS34-CII). The complex-specific subunits cause VPS34-CI to localize to autophagosome and VPS34-CII to endosome membranes. In mammalian cells, VPS34 complexes have four core subunits. Both complexes share the VPS34, VPS15, and BECLIN1 subunits. In addition, VPS34-CI has ATG14L, while VPS34-CII has UVRAG. In yeast, the

homologous complexes have common subunits known as Vps34, Vps15, and Vps30, while the unique subunits are Atg14 and Vps38 (Itakura *et al.*, 2008 [↗](#); Kihara *et al.*, 2001 [↗](#)). VPS34-CI and VPS34-CII have been studied by X-ray crystallography and cryo-EM. These studies revealed that the overall structures of both complexes are similarly V- or Y-shaped, with one arm (the kinase arm) composed of most of the VPS34 and VPS15 subunits and the other arm (adaptor arm) consisting of BECLIN1 with ATG14L in VPS34-CI and BECLIN1 with UVRAG in VPS34-CII (Baskaran *et al.*, 2014 [↗](#); Cook *et al.*, 2025 [↗](#)) (Ma *et al.*, 2017 [↗](#); Rostislavleva *et al.*, 2015 [↗](#); Tremel *et al.*, 2021 [↗](#); Young *et al.*, 2016 [↗](#); Young *et al.*, 2019 [↗](#)).

We recently found that organelle-specific recruitment of VPS34 complexes is achieved by RAB GTPases: ER/autophagosome-localized RAB1A interacts with and activates VPS34-CI, while early endosome-localized RAB5A interacts with and activates VPS34-CII (Tremel *et al.*, 2021 [↗](#)). The ‘Switch’ regions in the RAB proteins undergo substantial conformational changes when switching from GDP-to GTP-bound states (Pylypenko *et al.*, 2018 [↗](#)). RAB effectors interact with these regions and selectively recognize RABs in a nucleotide-dependent manner (Grosshans *et al.*, 2006 [↗](#)). A triad of invariant hydrophobic residues F, W, and Y located in Switch 1, Interswitch, and Switch 2 regions, respectively, can adopt different conformations to specifically bind effectors (Merithew *et al.*, 2001 [↗](#)). While VPS34-CII will bind soluble RAB5 that is not coupled to membranes, this binding has no influence on the VPS34 activity (Tremel *et al.*, 2021 [↗](#)). To activate VPS34-CII, RAB5 must be coupled to membranes containing PI3P (Tremel *et al.*, 2021 [↗](#)). VPS34 complexes will phosphorylate soluble lipids, *e.g.*, diC8PI, but their activities on membranes are greatly dependent on the charge and lipid saturation of the lipid membrane (Ohashi *et al.*, 2020 [↗](#)). Single-molecule kinetic studies are consistent with RAB5A activating VPS34-CII by both recruitment and allosteric activation (Buckles *et al.*, 2020 [↗](#)).

Our tomographic reconstruction of VPS34-CII on membranes provided a framework for understanding the mechanisms of VPS34-CII activation by RAB5A (Tremel *et al.*, 2021 [↗](#)). However, to improve the resolution of VPS34-CII, relative to the tomographic reconstruction, we have here determined high-resolution single-particle cryo-EM structures of VPS34-CII bound to RAB5A. Remarkably, we find that VPS34-CII-RAB5A complex has two independent RAB5A binding sites: one on VPS34 and one on the helical solenoid of VPS15. However, only the VPS34 site was detected in the previous tomographic reconstruction. This VPS34 site has three walls (a tripartite binding site) made of the VPS15 small globular domain (SGD, residues 723-805), the VPS15 WD40 (residues 968-1358), and the helical hairpin insertion in the VPS34 C2 domain (VPS34 C2HH, residues 170-210) (Cook *et al.*, 2025 [↗](#); Tremel *et al.*, 2021 [↗](#)). We previously attempted to eliminate RAB binding to VPS34 complexes by structure-guided mutation of the 199-REIE-202 stretch in VPS34 C2 HH helix 2 to AAAA (REIE>AAAA). VPS34-CI REIE>AAAA could neither bind nor be activated by RAB1A, while, surprisingly, the same mutation in VPS34-CII increased affinity for and activation by RAB5A (Tremel *et al.*, 2021 [↗](#)). The starkly different effects of this mutation on the two complexes remained unexplained by the cryo-ET study of VPS34-CII but can be explained by the new high-resolution single-particle cryo-EM structures.

Results

High resolution structure of a mutant VPS34-CII bound to RAB5A

Using a confocal microscopy-based bead-binding assay, we showed that mCherry-tagged VPS34-CII bound specifically to RAB5A (Figs. 1A [↗](#) and B, Figs. S1A [↗](#) to S1C [↗](#)). Interaction of VPS34-CII with RAB1A was too weak to measure (Fig. S1D [↗](#)). Consistent with our previous activity assays, the VPS34-CII REIE>AAAA mutation in the VPS34 C2 domain bound RAB5A about three times more tightly than the WT complex (Figs. 1B [↗](#) and S1E [↗](#)) (Tremel *et al.*, 2021 [↗](#)). Because the VPS34-CII REIE>AAAA mutant bound more tightly to RAB5A than WT VPS34-CII (Fig. 1B [↗](#)), we used it for our single-particle cryo-EM sample of the VPS34-CII-RAB5 assembly, then built and refined a structure at a 3.2 Å resolution (Figs. S2A, B [↗](#)). The structure of the mutant bound to RAB5A is remarkably similar to VPS34-CII-BATS-RAB5A on membranes that we determined previously at low resolution by cryo-ET (Tremel *et al.*, 2021 [↗](#)). There were only small changes in kinase and adaptor arms of the complex (Fig. S2C [↗](#)). The VPS15 helical solenoid was the best resolved region

of the complex, while N- and C-terminal extremities of the BECLIN1 and UVRAG subunits were less well resolved. RAB5A bound to a tripartite site composed of VPS15 SGD, VPS15 WD40 and VPS34 C2HH (Figs. S2D [and E](#)), which is consistent with our previous results (Tremel *et al.*, 2021 [and](#)). We used this mutant complex as an initial model for building the structure of wild type VPS34-CII-RAB5A (see below). One surprising feature of the mutant reconstruction was that there was weaker density that resembled a second molecule of RAB5A bound to the helical solenoid of VPS15 (Fig. S2A [and](#)). A model was built into this RAB5 density at partial occupancy, but this second site was even more apparent in the structure of a mutant in which the VPS34 RAB5 site was mutated (see next section).

Structures of RAB5A-binding mutants suggest that VPS34-CII has an unexpected second RAB5A binding site on the VPS15 subunit

To examine the structural contribution of the VPS34 C2 HH to RAB5A binding, we introduced a reverse charge mutation in VPS34, 199-REIE-202 to 199-ERIR-202 (REIE>ERIR), to disrupt interactions between RAB5A Switch 1 and VPS34 C2 HH helix 2. Interestingly, the GUV kinase assay showed that although the reverse charge mutation did not eliminate VPS34-CII activation by RAB5A, the extent of activation was reduced two-fold compared to wild type VPS34-CII (Fig. 1C [and](#)). Compared to wild type VPS34-CII, the reverse charge REIE>ERIR mutant VPS34-CII had a RAB5A binding affinity that was drastically reduced, to a K_D of $27 \pm 8.3 \mu\text{M}$ (Fig. 1D [and](#) Fig. S1F [and](#)). To investigate whether the partial reduction in VPS34-CII REIE>ERIR activation by RAB5A was due to incomplete disruption of the VPS34-RAB5A interaction, we determined the structure of a VPS34-CII reverse charge mutant in the presence of RAB5A (Fig. 1E [and](#)). As expected, we did not observe any density in the tripartite RAB binding site. However, consistent with the weak density on the VPS15 subunit in our structure of VPS34-CII with the VPS34 REIE>AAAA mutation, we now saw strong density for RAB5A bound to the VPS15 subunit of VPS34-CII REIE>ERIR. We fit and refined an atomic model for RAB5A-GTP into the new density (Fig. 1F [and](#)) on the VPS15 helical solenoid domain, with an interaction centered on the helix spanning VPS15 residues 570-585. RAB5A mediates this interaction mostly through its hydrophobic triad residues F57, W74, and Y89 (Fig. 1G [and](#)). The Interswitch residue W74 contacts VPS15 residues S579 and H580, as well as the Switch 1 residue F57. In turn, F57 also interacts with VPS15 H580, and its nearby residue T583. The Switch 2 residue Y89 is not in a direct contact with any VPS15 residues and instead binds to other RAB5A residues. This contrasts with the RAB5A binding to VPS34 C2 HH, where Switch 1 F57 has a stabilizing role for the switch, but does not make direct interaction with the effector-binding site. It is evident that RAB5A has generous plasticity in its hydrophobic triad region that allows binding to different molecular surfaces.

To verify that RAB5A binding the VPS15 site was not an artefact of cryo-EM sample preparation, we used hydrogen – deuterium exchange mass spectrometry (HDX-MS) and examined RAB5A binding to the wild-type VPS34-CII in solution (Fig. S3A [and](#)). We observed strong protection upon RAB5A binding not only in the VPS34 C2 HH helix 2 (Fig. S3B [and](#)), but also in the VPS15 helical solenoid domain, including helix 570-585 (Fig. S3C [and](#)). Protection in the VPS34 RAB5A binding site is consistent with our previous HDX-MS results (Tremel *et al.*, 2021 [and](#)), and protection on the VPS15 helical solenoid residues 570-585 indicates that RAB5A binds the VPS15 site both in solution and in the cryo-EM sample.

Wild type VPS34-CII can bind two copies of RAB5A simultaneously

We next determined the structure of wild-type VPS34-CII-RAB5A to further confirm the presence of the new RAB5A binding site on VPS15. After several rounds of heterogeneous refinement and 3D classification, we found that 30% of particles were apo VPS34-CII, 23% of particles had RAB5A bound to VPS34 only (VPS34-CII-RAB5A_{VPS34}), 32% of particles had RAB5A bound to VPS15-only (VPS34-CII-RAB5A_{VPS15}), and 15% of particles had RAB5A bound to both VPS34 and VPS15 (VPS34-CII-(RAB5A)₂) (Fig. 1H [and](#)). We subjected these particles to numerous rounds of focused 3D classifications to ensure that the density of VPS34-CII-(RAB5A)₂ originated from particles having two bound RAB5As and was not an artefact of combining VPS34-CII-RAB5A_{VPS15} particles with

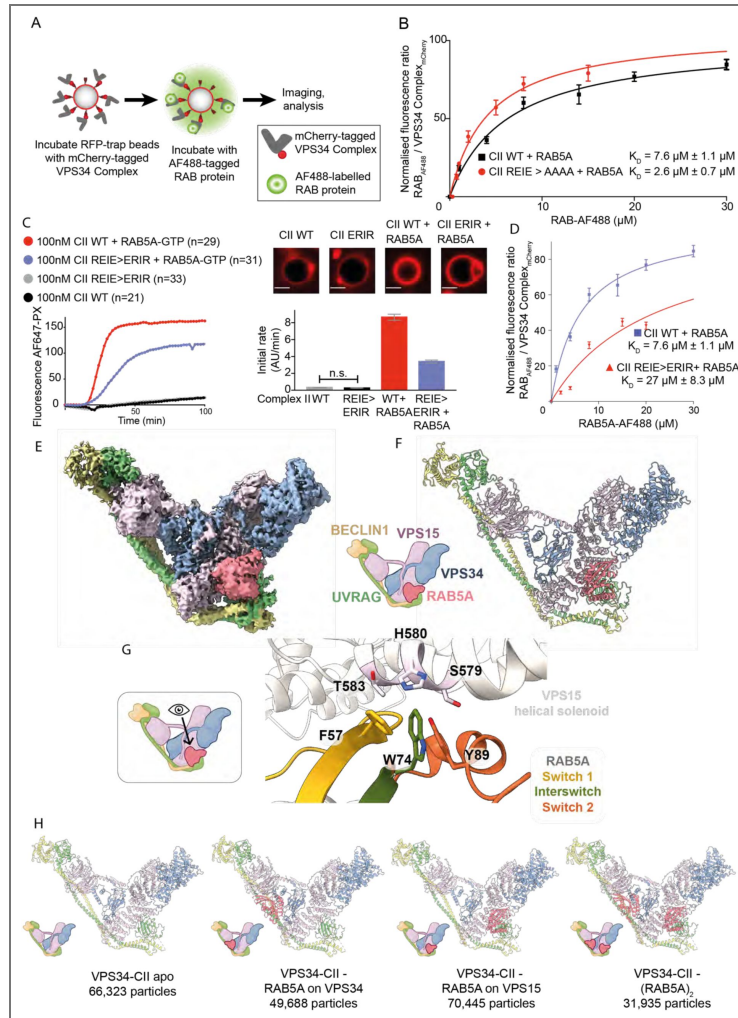


Figure 1. A Cryo-EM structure of human VPS34-CII with RAB5A showed a second RAB5 binding site on VPS15.

A. Design of VPS34 CII – RAB5 binding affinity measurement assay. RFP-trap beads were coated with mCherry-tagged VPS34-CII, incubated with various concentrations of AF488-labelled RAB protein and imaged by confocal microscopy. **B.** Quantification of RAB5-AF488 binding to VPS34-CII. Fluorescence intensity in the AF488 channel was normalised by the fluorescence intensity of mCherry channel. Plots are representative of three independent replicates for each experiment, and the reported K_d is average $\pm$ SD. Error bars in the plots are mean $\pm$ SD of the representative experiment. We verified that mCherry-tagging does not structurally compromise the VPS34 complexes (Fig. S1A) and that mCherry alone does not interact with RAB proteins (Fig. S1B). **C-H.** VPS34-CII has two RAB5-binding sites, one on VPS34 and the other on VPS15. The VPS34-CII REIE>ERIR reverse charge mutation eliminates RAB5A binding to the VPS34 site, but not the VPS15 RAB5-binding site. **C.** GU-based assay with the wild-type VPS34-CII and VPS34-CII carrying a REIE>ERIR mutation in VPS34 C2 HH, in the presence and absence of membrane-tethered RAB5A. The REIE>ERIR mutation partially reduces VPS34-CII activation by RAB5A. Reaction progress curves are shown on the left, and initial rates on the right as bar graphs. All means are significantly different with $p < 0.05$, except where not significantly different means are indicated by 'n.s.'. Experiments were performed in triplicate; error bars show $\pm$ SD of the representative experiment. Micrographs show the AF647-PX signal at the end of reaction, scale bar = 5 μ m. **D.** The REIE>ERIR mutation reduces RAB5A binding affinity compared to the wild-type VPS34-CII. The plot shows binding curves of RAB5A and VPS34-CII REIE>ERIR mutant together with the wild-type VPS34-CII + RAB5A binding curves. **E.** Composite cryo-EM map of VPS34-CII REIE>ERIR with RAB5A bound to VPS15 subunit. Icons of VPS34-CII in E-H are based on cartoons from BioRender. **F.** Cartoon representation of the atomic model for VPS34-CII REIE>ERIR-RAB5A. **G.** A close-up view of the RAB5A binding site on VPS15. The RAB5A hydrophobic triad engages VPS15 helical solenoid domain. RAB5A residues F57 (yellow) and W74 (green) interact with VPS15 residue H580, RAB5A Y89 (orange) contacts VPS15 residues S579 and H580. **H.** Atomic models of four different VPS34-CII states obtained from a cryo-EM sample of the WT VPS34-CII + RAB5A. Number of particles used for reconstruction of each state is given below the models. VPS34-CII was observed in apo state, with RAB5A bound to VPS34 C2 HH, with RAB5A bound to VPS15, or with RAB5A molecules bound to each site at the same K_m.

VPS34-CII-RAB5A_{VPS34} particles. Apart from the binding of two RAB5As, the structure of VPS34-CII-(RAB5A)₂ did not show significant conformational differences from the other RAB5-bound VPS34-CII structures, nor from the apo form. This indicates that VPS34-CII can interact with two RAB5A GTPases simultaneously, via two independent binding sites. The structure of the VPS34 REIE>AAAA mutant that bound RAB5 tighter than the WT VPS34 also has a second RAB5 binding site on VPS15, however, it is likely that the population of complexes in the cryo-EM sample contained both particles with and without RAB5A bound to VPS15, since the average density in the VPS15 RAB5 binding site was only about 40% of the average density in the VPS34 RAB5 binding site. Models were built and refined for all RAB5-associated VPS34-CII assemblies.

To test whether the RAB5A binding site on VPS15 has a role on VPS34-CII activation, we mutated the VPS15 579-SHMIT-583 stretch into 579-DDMIE-583, with the expectation that the charged, bulkier aspartic and glutamic acid residues would disrupt the RAB5A hydrophobic triad interaction with VPS15. As expected, the mCherry-tagged VPS34-CII VPS15 mutant could be activated by RAB5A but had about two-fold less activity than the wild-type VPS34-CII (Fig. 2A). The VPS34-CII VPS15 mutant had a K_D of $6.0 \pm 0.1 \mu\text{M}$ for RAB5A (Fig. 2B), which is comparable to wild type VPS34-CII, where $K_D = 7.4 \pm 1.1 \mu\text{M}$. To completely abolish RAB5A binding and activation of VPS34-CII, we combined the VPS34 REIE>ERIR and VPS15 SHMIT>DDMIE mutations. We verified that all the single and double mutants were structurally intact by nanoDSF (Fig. S4). The VPS15 + VPS34 double mutant VPS34-CII could not be activated by RAB5A (Fig. 2C), and we observed no binding of RAB5A to the double mutant complex II – coated beads (Fig. 2D). When the RAB5 binding sites in VPS34 and VPS15 are aligned with various species, the VPS34 RAB5-binding sequence REIE is not conserved in yeast *S. cerevisiae* and *S. pombe*, where it is rather reverse-charged (Fig. 2E). In contrast, the VPS15 RAB5-binding sequences are highly conserved through evolution. Indeed, the yeast Vps15 RAB5 non-binding mutant (*S. cerevisiae* 622-SHLITY-627 > DDLIEY) in yeast VPS34-CII was sufficient to abolish the binding to the yeast RAB5 orthologue, Vps21 *in vitro* (Fig. S5). Also, the yeast Vps15 SHLITY>DDLIEY mutant did not show strong colocalization with Vps21 (Fig. 2F), and the same mutant is defective in endocytic sorting measured by CPY processing (Fig. 2G). Taken together, these results show that RAB5A binds to two sites on human VPS34-CII, both sites are required for full activation, and that two RAB5A GTPases can bind VPS34-CII at the same time. It is likely that RAB5 binding in eukaryotic VPS34-CII originated at the VPS15 site, with the second binding site on VPS34 having been acquired during evolution.

RAB5A binding to VPS34 site does not induce conformational changes in VPS34-CII

As described above, the cryo-EM sample of wild-type VPS34-CII-RAB5A contained both the apo and RAB5A-bound forms of VPS34-CII, which enabled us to confidently examine whether binding of RAB5 affected the conformation of the VPS34-CII. We could see that similarly to what was reported for RAB1A binding to VPS34-CI (Cook *et al.*, 2025), RAB5A binding to VPS34-CII did not induce large conformational differences (Fig. 3A and S6A,B,C). Furthermore, the WT VPS34-CII-RAB5A structure did not differ greatly from VPS34-CII REIE>AAAA-RAB5A, nor a reinterpreted VPS34-CII-BATS-RAB5A model based on our cryo-ET data previously reported (Tremel *et al.*, 2021) (Figs. S6D and E).

RAB specificity for VPS34 complexes arises from RAB Switch regions and from the VPS34 C2 HH conformation

We compared a previously reported VPS34-CI bound to RAB1A (VPS34-CI-RAB1A) (Cook *et al.*, 2025) with the VPS34-CII-RAB5A structures (Fig. 3). Both RABs bind VPS34 complexes in the same tripartite site with similar orientations (Fig. 3A). The most prominent element of the RAB binding site is the VPS34 C2 HH. In VPS34-CI, both helices of the C2 HH engage RAB1A (Fig. 3B, left), while in VPS34-CII, only one VPS34 C2 HH helix engages RAB5A (Fig. 3B, right). Also, in VPS34-CI, the VPS15 WD40 domain and RAB1A are separated by 27 Å (Fig. 2B, left), while in VPS34-CII the distance between equivalent points in RAB5A and WD40 is only 9 Å (Fig. 3B,



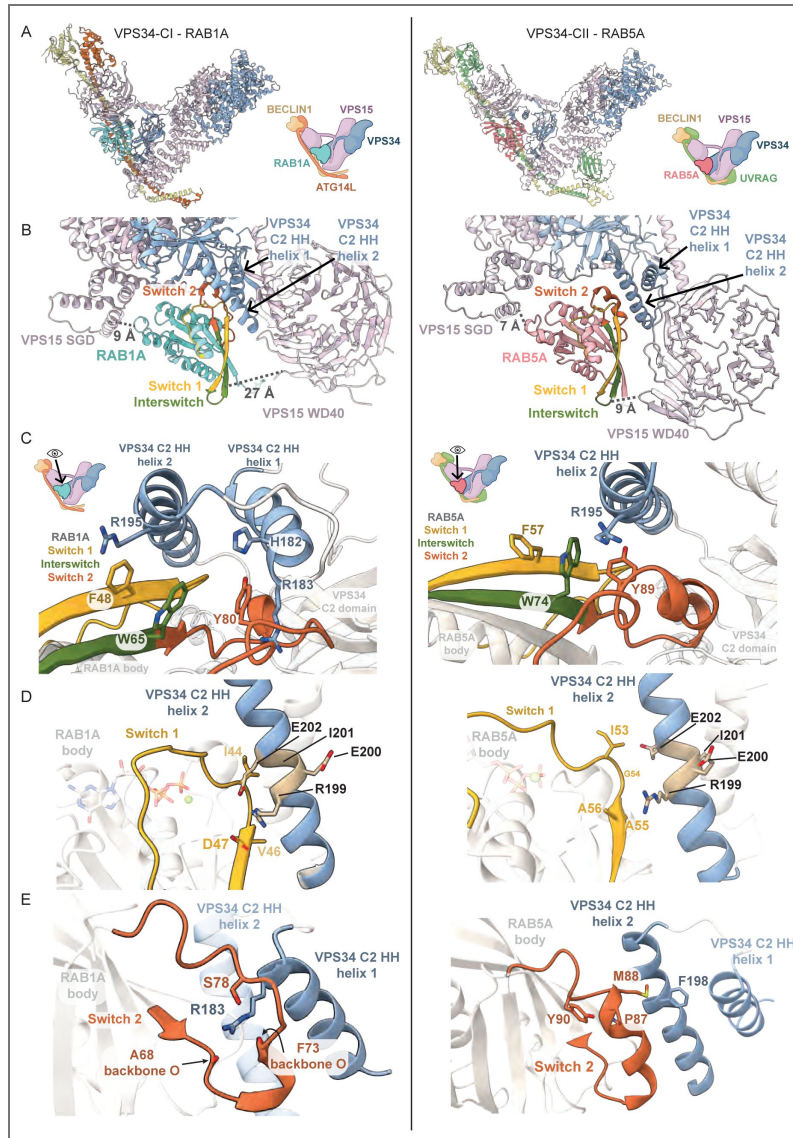


Figure 3. Comparison of VPS34-CI-RAB1A and VPS34-CII-RAB5A binding modes.

Left panels show features of VPS34-CI-RAB1A (PDB: 9MHG). Right panels show features of VPS34-CII-RAB5A. Icons of VPS34 complexes are based on cartoons from *BioRender*. **A.** RAB1A and RAB5A bind to the equivalent site in both complexes. **B.** RAB1A binds both helices of the VPS34 C2 HH. Switch 1 region (yellow) binds helix 2, while Switch 2 (orange) binds helix 1. RAB5A contacts only helix 2 from VPS34 C2 HH. The 9 Å distance between RAB1A and VPS15 SGD is comparable to the 7 Å distance between RAB5A and VPS15 SGD. The VPS15 WD40-RAB distance is larger in VPS34-CI-RAB1A (27 Å) than in VPS34-CII-RAB5A (9 Å). Distances were measured between the α -carbon atoms in selected VPS15 and RAB residues: VPS15 S776 and RAB1A T130 / RAB5A N139 for RAB-SGD distance; VPS15 G1298 and RAB1A T59 / RAB5A T68 for RAB-WD40 distance. The RAB residues are equivalent according to MSA. **C.** Hydrophobic triad residues in RAB1A and RAB5A bind VPS34 C2 HH in different modes. In RAB1A, residue F (yellow) contacts C2 HH helix 2, residue Y (orange) engages C2 HH helix 1, and residue W (green) is not involved in direct contact with the C2 HH. By contrast, in RAB5A, it is residue F (yellow) that is not involved in binding C2 HH, while both residues W (green) and Y (orange) engage the same VPS34 C2 HH helix 2. VPS34 C2 HH helix 2 residue R195 interacts with hydrophobic triad residues in both RAB1A and RAB5A. VPS34 residue R183 in helix 1 is involved in binding RAB1A, but not RAB5A. **D.** Switch 1 region (yellow) interacts with VPS34 C2 HH helix 1 in both VPS34 complex-RAB assemblies. RAB1A residue D47 forms a salt bridge with VPS34 residue R199. RAB5A Switch 1 has limited contact with VPS34 C2 HH helix 1. **E.** Switch 2 region (orange) adopts different conformations in VPS34-CI-RAB1A and VPS34-CII-RAB5A. In RAB1A, Switch 2 is in a loop conformation, embracing VPS34 residue R183 in VPS34 C2HH helix 1 that in return stabilizes the loop via polar contacts to RAB1A S78 and the backbone carbonyls of A68 and F73. In RAB5A, Switch 2 is in a helical conformation potentially stabilised by a hydrophobic interaction between RAB5A residues Y90 and P87. Switch 2 makes minimal contact with C2HH helix 2 through residue M88 that interacts with F198 in VPS34.

right). These differences in the tripartite binding pocket are not due to RAB binding, since WT apo VPS34-CII has the same overall conformation as the RAB5-bound conformation (Fig. S6C). Instead, the difference in the binding site appears to arise from the influence of the UVRAG subunit in VPS34-CII instead of the ATG14L subunit of the VPS34-CI. The more open RAB pocket adjacent to the VPS15 WD40 in VPS34-CI suggests that the WD40 might make a larger contribution to RAB binding in complex VPS34-CII than in complex VPS34-CI, indicating that difference in the geometry of the tripartite site in the two complexes is the most important contribution to the RAB specificity.

In the VPS34 RAB-binding site, there are considerable differences between VPS34-CI and VPS34-CII in the interaction between the VPS34 C2 HH and the RAB hydrophobic triad, which is made of invariant hydrophobic residues F, W, and Y located in the Switch 1, Interswitch, and Switch 2 regions (Fig. 3C). While the same elements are involved in both RAB-VPS34 complexes, they are arranged differently. In RAB1A (Fig. 3C, left), Switch 1 residue F48 forms a cation- π stacking interaction with residue R195 in VPS34 C2 HH helix 2. Residue Y80 in Switch 2 interacts with R183 in VPS34 C2 HH helix 1. The Interswitch residue W65 does not engage VPS34 C2 HH, but instead contacts RAB1A residue F48 and others (Fig. S7A). In contrast, for the VPS34-CII-RAB5A structure (Fig. 3C, right), Interswitch residue W74 forms a cation- π stacking interaction with the VPS34 R195. In this complex, VPS34 R195 is also engaged by Y89 in the RAB5A Switch 2. RAB5A Switch 1 residue F57 does not directly participate in VPS34 binding and instead forms an intramolecular π - π stacking interaction with RAB5A W74.

In both RAB1A and RAB5A, the Switch 1 region is close to VPS34 C2 HH helix 2 (Fig. 3D). In VPS34-CI-RAB1A, Switch 1 residue D47 forms a salt bridge with VPS34 R199, which, in turn, interacts with VPS34 E202 (Fig. 3D, left). The importance of the VPS34 R199 - RAB1A D47 interaction is supported by our previous finding that the VPS34 REIE>AAAA mutation abolishes VPS34-CI recruitment and activation by RAB1A (Tremel *et al.*, 2021). In VPS34-CII-RAB5A, Switch 1 of RAB5A is also positioned close to helix 2 of the VPS34 C2 HH (Fig. 3D, right). However, instead of potential side chain hydrogen bonds or salt links between these elements, RAB5A has a non-polar A56 residue in the position equivalent to RAB1A D47 (Fig. S8), and it is bordered by aliphatic residues I53, G54, and A55. This aliphatic RAB5A Switch 1 region interacts more favorably with the non-polar VPS34 AAAA mutant than with the polar REIE motif in WT VPS34 C2 HH helix 2 (Fig. S2E), resulting in increased activation of VPS34-CII REIE>AAAA by RAB5A.

The Switch 2 region in both RABs interacts with different VPS34 C2 HH helices (Fig. 3E). When bound to VPS34-CI, the flexible RAB1A Switch 2 loops around the positively charged R183 in VPS34 C2 HH helix 1, forming polar contacts with it via residue S78 and the backbone carbonyl oxygens of A68 and F73 (Fig. 3E, left), with distances ranging from 2.8 Å to 3.4 Å, suggesting potential hydrogen bonding (Fig. S7B). This conformation of Switch 2 contrasts with a more helical conformation in other RAB1A-effector complexes (Cheng *et al.*, 2012; Dong *et al.*, 2012; Mishra *et al.*, 2013). In a complex with VPS34-CII, RAB5A Switch 2 adopts a compact helical conformation stabilized by an intramolecular interaction between residues P87 and Y90. The M88 residue in this helical Switch 2 makes a non-polar interaction with F198 in VPS34 C2 HH helix 2 (Fig. 3E, right). Interestingly, RAB1A S78 and RAB5A P87 share the same position in the multiple sequence alignment (Fig. S8), suggesting that the function of this position is to stabilize a RAB-specific conformation of the Switch 2 region.

Overall, to bind VPS34-CII, RAB5A uses two interfaces (Switch 1 and the hydrophobic triad) and relies on the tripartite site in VPS34-CII. In contrast, RAB1A binding to VPS34-CI involves three interfaces (Switch 1, Switch 2, and the hydrophobic triad) that make polar and nonpolar interactions with both VPS34 C2 HH helices, but does not appear to contact other elements of the VPS34-CI tripartite site.

VPS34-CII has a zinc finger at the N-terminus of BECLIN1

The most obvious characteristic of higher-resolution VPS34-CII that we observed was the presence of an additional RAB5-binding site on VPS15, near the base of the V-shaped complex. A second surprising feature of our higher-resolution structures of human VPS34-CII were additional

features at the base of the V-shaped complex, including the N-terminal end of BECLIN1 that previously were not apparent. While there is no indication that these BECLIN1 features are related to RAB binding, they are likely to be important for the function of VPS34-CII.

The BECLIN1 N-terminal region has two highly conserved CXXC motifs (18-CXXC-21 and 137-CXXC-140) flanked by an intrinsically disordered region (IDR) (Mukhopadhyay *et al.*, 2024 [↗](#)), which has hindered detailed studies of this region (Chen *et al.*, 2023 [↗](#); Rostislavleva *et al.*, 2015 [↗](#); Young *et al.*, 2019 [↗](#)). Our cryo-EM densities allowed us to dock and refine atomic models of the BECLIN1 N-terminus predicted by AlphaFold3 (Abramson *et al.*, 2024 [↗](#)). In VPS34-CII there is a Zn^{2+} coordinated by the two BECLIN1 CXXC motifs to form a zinc finger (Fig. S9 [↗](#)). In addition, the N-terminal C2 domain of UVRAG could be clearly built. The structures show that the C2 domain is sandwiched between the N-terminal zinc finger of BECLIN1 and the long helical hairpin in the middle of the VPS15 solenoid (residues 465-545) (Fig. 4 [↗](#)). There was a density at the edge of strand $\beta 2$ of the UVRAG C2 domain that appeared to augment the second β -sheet of the C2 domain. The AlphaFold3 model also had an augmented β -sheet and we have attributed this augmenting strand to BECLIN1 residues 78-83. This mostly ordered and relatively compact N-terminal region of BECLIN1 and UVRAG contrasts with the analogous region in VPS34-CI that was almost completely disordered (Cook *et al.*, 2025 [↗](#)). The N-terminal region of BECLIN1 also contains the BH3 motif of BECLIN1 (residues 105-130) to which Bcl-2 family proteins can bind to inhibit autophagy. It is not immediately clear how Bcl-2 binding would inhibit the VPS34 complexes, but the BH3 forms one of the two contacts between the UVRAG C2 domain and other subunits of the VPS34-CII (purple helix in Fig. 4 [↗](#)).

Discussion

Our single-particle cryo-EM structures of assemblies of human VPS34-CII-RAB5A showed unanticipated aspects of VPS34 regulation. The human VPS34-CII VPS34 REIE>AAAA mutant bound RAB5A with a greater affinity than WT VPS34-CII and enabled the highest resolution VPS34-CII structure (3.2 Å). We have also determined apo structures of wildtype VPS34-CII at a resolution of 3.9 Å. These structures allowed us to delineate the molecular basis for RAB specificity of VPS34 complexes and determine that the differences in the overall shape of the VPS34 RAB binding pocket is not a consequence of RAB binding but a consequence of the complex-specific subunit (UVRAG versus ATG14L). We previously observed that VPS34-CII was not activated by RAB1 but that VPS34-CI is weakly activated by RAB5 *in vitro* (Tremel *et al.*, 2021 [↗](#)). This differential selectivity may arise from the tripartite RAB5 binding site in VPS34-CII being narrower than the equivalent site in VPS34-CI. This may restrict the geometry of RAB binding in a way that would only allow RAB5A, but not RAB1A, to fit and bind the tripartite site of VPS34-CII.

One subunit difference between the two apo complexes causes conformational changes in the VPS34 tripartite RAB-binding site. Most importantly, in VPS34-CI, the C2 HH is positioned with both helices available for RAB1A, while in VPS34-CII only one helix from the C2 HH is accessible for RAB5A binding. The diminished interaction with the C2 HH in the VPS34-CII-RAB5A assembly appears to be compensated by a narrowing of the tripartite RAB5A pocket with additional, non-switch interactions between RAB5A and VPS15. Remarkably, the complex-specific ATG14L and UVRAG subunits do not come in direct contact with RABs. Given that the apo-VPS34-CII closely resembles the VPS34-CII-RAB5A, it appears that ATG14L tunes binding to RAB1A by twisting the common subunits VPS34, VPS15 and BECLIN1 into a RAB1-specific conformation.

RAB1A and RAB5A bind VPS34-CI and -CII via the Switch and Interswitch regions. In both RAB-VPS34 complex assemblies, the RAB hydrophobic triad is positioned at the linker between the VPS34 C2 HH helices and engages it in a similar manner. In contrast, the rest of the RAB Switch regions make VPS34 complex-specific interactions. RAB1A Switch 1 residue D47 particularly stands out, as it forms a salt bridge with VPS34, while in RAB5A Switch 1 is mostly hydrophobic and forms entirely different contacts with VPS34. The hydrophobic triad likely provides most of the binding energy, and the Switch regions confer specificity, as has been observed with other RAB-effector complexes (Lucas *et al.*, 2014 [↗](#)). The distinctive, hydrophobic RAB5A Switch 1 region interaction



Upper: The N-terminal region of BECLIN1 (blue) is stabilized by a zinc finger (Zn²⁺ grey), and this region together with the long helical hairpin in the VPS15 solenoid region sandwich the UVRAG C2 domain. Two basic residues (green spheres) comprise a proposed PI3P-binding region (He *et al.*, 2013). RAB5 on VPS15 is indicated with an orange molecular surface. The BH3 helix of BECLIN1 is colored magenta. Lower: linear domain compositions of VPS15, BECLIN1, and UVRAG. The colors correspond to equivalent domains in the cartoon.

with the VPS34 C2 HH also explained why the VPS34-CII RAB5A binding site mutation REIE>AAAA, which was meant to inhibit RAB5A binding, increased the binding affinity and activation of VPS34-CII (Tremel *et al.*, 2021 [DOI](#)).

Our VPS34 complex structures showed an N-terminal VPS15 myristoylation and a GDP bound to the VPS15 kinase domain (Fig. S10 [DOI](#)). The myristoyl-binding pocket is in the vicinity of the interface with VPS34 (Fig. S10 [DOI](#)). It was proposed that VPS34 is activated when the myristoyl moiety is ejected from the VPS15 N-lobe and inserts into a lipid membrane, however, the same study also showed that a non-myristoylatable mutant (G2A) is fully functional (Cook *et al.*, 2025 [DOI](#)), which is consistent with a previous yeast Vps15 study (Herman *et al.*, 1991 [DOI](#)). Therefore, the precise role of the VPS15 myristoylation in activation remains to be elucidated. The myristoyl group inserting into membranes would increase association with membranes, but it also has been shown that RAB5 makes a very prominent contribution to the association with membranes (Buckles *et al.*, 2020 [DOI](#)).

What Cook *et al.* (Cook *et al.*, 2025 [DOI](#)) designated as the ‘inactive’ conformation (PDB ID 9MHG) resembles the conformation that we report here for our cryo-EM studies of VPS34-CII. In any of our datasets we did not observe the minor ‘active’ conformation previously reported (Cook *et al.*, 2025 [DOI](#)) which involved a 140° rotation of the VPS34 Helical-Catalytic (HELCAT) unit (PDB ID 9MHH). It may be that the VPS34-CII is poised for activation but not yet activated under the conditions in our cryo-EM study. It is also possible that the transition state on a membrane is sufficiently short-lived that we have not been able to capture it, even though we have RAB5 bound in the single-particle structures in the absence of membranes and RAB5 bound in the presence of membranes from our previous cryo-ET study (Tremel *et al.*, 2021 [DOI](#)).

Surprisingly, we discovered a new RAB5A binding site on the VPS15 subunit of VPS34-CII. Like the previously known VPS34 site and other RAB-effector interactions (Eathiraj *et al.*, 2005 [DOI](#); Hou *et al.*, 2011 [DOI](#); Lucas *et al.*, 2014 [DOI](#)), this newly discovered VPS15 site has micromolar-range affinity for VPS34-CII. In addition, the VPS15 RAB5 binding site is α -helical, which also is typical in RAB effectors (Kawasaki *et al.*, 2005 [DOI](#); Rai *et al.*, 2019 [DOI](#)). The VPS15 site consists of a single α -helix while the VPS34 site has two helices.

Altogether the avidity of VPS34-CII for RAB5 is dictated by interactions in three disparate regions: helix 2 of the VPS34 helical insertion in the C2 domain, the VPS15 WD40 in the tripartite binding site, and the VPS15 solenoid interaction with separate RAB5. In contrast, VPS34-CI interacts with RAB1 using only a single region, the helical insertion in the VPS34 C2 domain. VPS34-CI appears to compensate for lack of the other two regions by interacting with both helices of the helical insertion in the VPS34 C2 domain. This is enabled by the unique bend of the ATG14L-containing adaptor arm that shifts the helical insertion to bring both helices into contact with bound RAB1.

In contrast to the VPS34 site, RAB5A binding to VPS15 does not involve extensive specific interactions through the Switch regions beyond the hydrophobic triad. This might account for the lower affinity of the VPS15 binding site compared to the VPS34 site. RAB1A does not bind or activate VPS34-CII, suggesting either that the conformation of the hydrophobic triad and the Switch regions in RAB1A are not complementary to VPS15 helix 570-585, or that the VPS15 RAB-binding site is blocked by other elements in VPS34-C1. RAB5 binding assays are consistent with a higher affinity VPS34 site and a lower affinity VPS15 site binding independently. However, functional assays indicate that each RAB5A binding site contributes to activation equally. The lack of synergy in the binding assays, despite clear synergy in activity assays is consistent with the complex binding to two RAB5A molecules on the same membrane. Such synergy could not be observed in the binding assays because of the way they were conducted: VPS34-CII was bound to beads and soluble RAB5A was titrated. To observe the synergy that we observe on membranes would require the RAB5 be fixed to beads, while titrating VPS34-CII, and this arrangement would be impossible because of the limited solubility of the VPS34-CII. How the two RAB5A binding sites in VPS34-CII are coordinated in mammals remains to be seen.

There are only a few other cases where multivalent RAB effectors contain binding sites with different affinities (Rai *et al.*, 2019 [DOI](#)). For example, MICAL1, a cytoskeleton regulatory protein, contains two RAB8 binding sites, one with high and one with low affinity for RAB8. The higher affinity site is used for initial MICAL1 recruitment to membranes, while the lower affinity site binds later (Rai *et al.*, 2019 [DOI](#)). The two RAB5A binding sites on VPS34-CII might be a part of a similar mechanism that may control VPS34 activity on early endosomes. VPS34-CII would be fully activated only when present on highly concentrated RAB5A membrane domains and in absence of other RAB5A effectors that might bind RAB5A more tightly than VPS34-CII VPS15 site. On the other hand, VPS34-CII may cooperate with RAB5A effectors EEA1, Rabaptin-5, and Rabenosyn-5 to maintain early endosomal RAB5A domains that are required for endosome tethering and fusion (Zerial & McBride, 2001 [DOI](#)). In addition, VPS34-CII could also be involved in vesicle tethering of two RAB5 proteins from different vesicles bound a single VPS34-CII. More sensitive experiments such as single molecule analysis will be required to investigate these possibilities. Finally, two RAB5A molecules might be required to correctly orient the VPS34 complex on the membrane, as both RAB binding sites are on the same side of the complex. While we did not observe two RAB5A proteins bound to VPS34-CII-BATS in the cryo-ET structure (Tremel *et al.*, 2021 [DOI](#)), the lower occupancy of RAB5A on vesicles used for the cryo-ET structure on membranes might have meant that the VPS15 RAB5A binding site was not occupied.

Interestingly, multiple sequence alignments suggest that yeast Vps34-CII has a single RAB5 binding site, and this would be on the Vps15 subunit. As with human RAB5, the yeast RAB5 orthologue Vps21 is required in endocytic trafficking (Singer-Kruger *et al.*, 1994 [DOI](#)), and the RAB1 homolog Ypt1 is involved in autophagy initiation (Yao *et al.*, 2024 [DOI](#)). Furthermore, Rab-pulldown experiments identified direct or indirect interactions with Vps15 (Graef *et al.*, 2013 [DOI](#); Nagano *et al.*, 2024 [DOI](#)). We showed that disrupting the Vps21 binding site in yeast Vps15 abolished the binding of Vps21 to yeast complex II *in vitro*, and inhibited endocytic sorting *in vivo*. It is possible that the RAB binding site on VPS15 evolved first, and the second RAB binding site on VPS34 evolved in higher organisms, as the complexity of membrane compartments increased the demand for tighter spatial and temporal regulation of PI3P production. It will be interesting to investigate how these RAB5A binding sites contribute to VPS34-CII function in human cells.

Methods

Lipid mixtures

ID	Composition	Descriptor	Purpose
STGUV2	18% liver PI, 10% DOPS, 17% DOPE, 55% DOPC, 0.03% DO-Liss-Rhod-PE, 0.03% DSPE-PEG(2000)-Biotin	DO base	For assays without RABs
STGUV10	18% liver PI, 10% DOPS, 7% DOPE, 10% DOPE-MCC, 55% DOPC, 0.03% DO-Liss-Rhod-PE, 0.03% DSPE-PEG(2000)-Biotin	DO base + 10% PE-MCC	For assays with RABs

Lipids

Lipid	Manufacturer	Catalog number
Liver PI (mixed chain, Bovine)	Avanti Polar Lipids, Inc.	840042C
DOPS	Avanti Polar Lipids, Inc.	840035C
DOPE	Avanti Polar Lipids, Inc.	850725C
DOPC	Avanti Polar Lipids, Inc.	850375C
DO-Liss-Rhod-PE	Avanti Polar Lipids, Inc.	810150C
DSPE-PEG(2000)-Biotin	Avanti Polar Lipids, Inc.	880129C
DOPE-MCC	Avanti Polar Lipids, Inc.	780201C

Plasmids

ID	Construct	Backbone	Marker	Use	Reference	addgene #
pJB78	GST-TEV-HsRAB1A Q70L C26S C126S, 1-204	pOPTG	amp	Cryo-EM, GUV assays	(Tremel <i>et al.</i> , 2021)	235738
pYO1005	HsUVRAG_FL, untagged	pCAG	amp	Bead assays	(Tremel <i>et al.</i> , 2021)	235702
pYO1015	HsVPS15_FL-3xTEV-ZZ	pCAG	amp	Cryo-EM, GUV assays	(Tremel <i>et al.</i> , 2021)	235704
pYO1025	HsVPS34_FL, untagged+hsVPS15-3xTEV-ZZ	pCAG	amp	Cryo-EM, GUV assays, Bead assays	(Ohashi <i>et al.</i> , 2016) (Ohashi <i>et al.</i> , 2020) (Tremel <i>et al.</i> , 2021) (Buckles <i>et al.</i> , 2020)	235705
pYO1031	HsBeclin 1_FL+HsUVRAG_FL, both untagged	pCAG	amp	Cryo-EM, GUV assays	(Tremel <i>et al.</i> , 2021)	235706
pYO1274	HsVPS34 (REIE to AAAA)_FL, untagged	pcdna4/TO	amp	Cryo-EM, GUV assays	(Tremel <i>et al.</i> , 2021)	235709
pYO1344	HsVPS34 R199E E200R E202R_FL, untagged	pcdna4/TO	amp	Cryo-EM, GUV assays	this study	235711

pYO1280	HsVPS34_FL WT-EGFP	pEEGF P-C1	km	Cell biology	(Tremel <i>et al.</i> , 2021)	235712
pYO1356	HsVPS34 R199E E200R E202R_FL-EGFP	pcdna4/TO	amp	Cell biology	this study	235713
pYO1846	mCherry-HsBeclin1_FL	pCAG	amp	Bead assays	this study	235714
pYO1847	HsVPS15 S579D_H580D_T583 E_FL-3x tev-ZZ	pCAG	amp	Cryo-EM, GUV assays	this study	235715
pYO1865	HsVPS15 S579D_H580D_T583 E_FL, untagged	pCAG	amp	Cell biology	this study	235716
pYO350	HsVPS15_FL, untagged	pcDNA4/TO	amp	Cell biology	(Tremel <i>et al.</i> , 2021)	235717
STp8 (pOP823)	HsRAB5A Q79L C19S,C63S, 1-212	pOPINS	kan	GUV assays, CryoEM	(Tremel <i>et al.</i> , 2021) (Buckles <i>et al.</i> , 2020)	235718
pYO1645	GST-PX domain with Cys at 2 and C85S	pOPTG	amp	GUV assays	(Buckles <i>et al.</i> , 2020)	235719
pYO634	ScVPS15(promoter+ WT FL coding)-EGFP	pRS416	amp	Yeast experiments	this study	235720
pYO1880	ScVPS15 (promoter+ S622D_H623D_T626 E_FL coding)-EGFP	pRS416	amp	Yeast experiments	this study	235721
pYO950	ScVPS15 (promoter + WT_FL coding)3xFLAG	pRS416	amp	Yeast experiments	this study	235722
pYO1877	Scvps15 (promoter + S622D_H623D_T626 E_FL coding) - 3xFLAG	pRS416	amp	Yeast experiments	this study	235723
pYO1510	mCherry-TEV-linker-14xhis	pOP	amp	Bead assays	this study	235739
pYO1006	HsBECLIN1 untagged	pCAG	amp	GUV assays	(Tremel <i>et al.</i> , 2021) (Ohashi <i>et al.</i> , 2020)	235724
pYO1879	PHO5 pro-dRFP-ScVPS21 Q66L-LEU2	pBluescript (KS-)	amp	Yeast experiments	this study	242480
pYO1919	ScVPS21 Q66L, 1-207A-EGFP	pOPTH	amp	Bead assays	this study	235769
pRS416	control, backbone vector	pRS416	amp	Yeast experiments	(Sikorski & Hieter, 1989)	GenBank: U03450.1

Protein purification

Human VPS34 Complex II

To prepare VPS34-CII, Expi293 suspension cells (ThermoFisher A14527) were grown in Expi293 Expression Medium (ThermoFisher A1435102) at 37°C, 8% CO₂, shaking at 125rpm. At density of 2×10⁶/L, cells were transfected with 1.1 mg DNA/L cells using polyethylenimine ‘MAX’ (Polysciences 24765, 1mg/mL in PBS) at 3 mg/L cells as the transfection reagent. Cells were grown further in the same conditions as above for 68 hours, then harvested by centrifugation at 3000 g for 15 min, flash frozen in liquid nitrogen, and stored at -80 °C until use.

To purify the protein, cells were resuspended in 50 mL/2 L cells of lysis buffer (50 mM HEPES pH 8.0, 300 mM NaCl, 1% Triton X-100 (Fisher BioReagents BP151), 12% glycerol, 2 mM MgCl₂, 1 mM phenylmethylsulfonyl fluoride (PMSF, Thermo Fisher Scientific 36978), 0.5 mM tris(2-carboxyethyl)phosphine (TCEP), and 1x EDTA-free protease inhibitor cocktail tablet (Roche 11873580001)) and incubated with slow stirring. Lysate was cleared by centrifugation at 142,000 g

for 30 min at 4 °C in a Type 45 Ti rotor (Beckman Coulter) and filtered through a 5 µm syringe filter (Sartorius 17594). Cleared lysate was incubated for 3.5 hours at 4 °C and 8 rpm rotation with 3 mL IgG beads (Cytiva 17-0969) equilibrated with lysis buffer. Beads were then transferred to a gravity column, washed with 50 mL ice-cold wash buffer (50 mM HEPES pH 8.0, 300 mM NaCl, 0.1% Triton X-100, 12% glycerol, 50 mM MgCl₂, 0.5 mM TCEP, 4 µg/mL bovine pancreatic RNase (Sigma 83834)), and incubated with 50 mL wash buffer supplemented with 5 mM ATP for 20 minutes. Beads were further washed with another 50 mL wash buffer and 100 mL TEV buffer (50 mM HEPES pH 8.0, 300 mM NaCl, 0.5 mM TCEP), then resuspended in 10 mL TEV buffer and incubated with 40 µL of 4.4 mg/mL TEV protease (house made) overnight without rotation to remove the C-terminal ZZ-tag on VPS15. Flow-through was collected and remaining protein eluted three times with 5 mL TEV buffer followed by 7 mL TEV buffer supplemented with 0.1% CHAPS (Calbiochem 220201). Eluted fractions were combined, passed through a 0.22 µm syringe filter (Elkay E25-PS22-50S), concentrated to 0.5 mL using a 100k MWCO concentrator (Millipore UFC810024), and injected onto an S200 10/300 GL gel filtration column equilibrated with 20 mM HEPES pH 8.0, 300 mM NaCl, 0.5 mM TCEP. Eluted fractions were inspected by SDS-PAGE, selected fractions were pooled, concentrated, frozen in liquid nitrogen in 2 µL aliquots, and stored at -80°C until use. Mutant VPS34 complexes and complexes with an N-terminal mCherry tag on BECLIN1 were purified in the same way as wild type complexes.

p40^{phox} PX domain purification and labelling

To prepare the p40^{phox} PX domain, plasmid pYO1645 was transformed into *E.coli* OverExpress™ C41(DE3) chemically competent cells (Lucigen 60442). Transformed bacteria were cultured in 6 L 2xTY medium supplemented with 0.1 mg/mL ampicillin (Formedium AMP50) at 37 °C until OD₆₀₀=0.8, then expression was induced with 0.5 mM IPTG, and bacteria were incubated for further 19 hours at 18 °C. Cells were pelleted by centrifugation at 6700 g for 25 minutes and pellets were resuspended in 100 mL lysis buffer (50 mM HEPES pH 8.0, 150 mM NaCl, 12% glycerol, 10 mM DTT, 1 mM PMSF, 1x EDTA-free protease inhibitor cocktail tablet). Cells were sonicated for 5 min on ice (2 s on, 3 s off, 60% amplitude), and lysate cleared by centrifugation at 142,000 g for 15 min at 4 °C in a Type 45 Ti rotor. Supernatant was filtered through a 5 µm syringe filter and incubated with 4 mL GS4B-GST beads (Cytiva 17-0756) equilibrated in lysis buffer for 1 hour at 4 °C at 8 rpm rotation. Beads were then transferred to a gravity flow column, washed with 50 mL lysis buffer, 200 mL wash buffer (50 mM HEPES pH 8.0, 300 mM NaCl, 12% glycerol, 10 mM DTT), and 50 mL

TEV buffer (50 mM HEPES pH 8.0, 300 mM NaCl, 2 mM TCEP). Beads were resuspended in 10 mL TEV buffer, then 80µL of TEV protease at 4.4 mg/mL was added, and sample were incubated overnight without rotation to cleave the N-terminal GST-tag. Flow-through and four 5 mL elutions with TEV buffer were pooled and supplemented with 20 mM imidazole, then filtered through a 0.22 µm syringe filter and incubated with 0.5 mL Ni-NTA agarose beads (Qiagen 1018244) equilibrated with TEV buffer for 45 minutes at 4 °C at 8 rpm rotation to remove the His₆-tagged TEV protease. Ni-NTA beads were transferred to a gravity flow column and the collected flow-through was again filtered through a 0.22 µm syringe filter and supplemented with 5 mM TCEP. Protein was concentrated to 1 mL using a 10k MWCO concentrator (Millipore UFC901024) and injected onto an S75 16/60 gel filtration column equilibrated in 20 mM HEPES pH 8.0, 150 mM NaCl, 1 mM TCEP. The peak fractions were pooled and concentrated to 31.5 mg/mL.

To label the p40^{phox} PX domain, 1 mg of AlexaFluor 647 C2 maleimide (Invitrogen A20347) was dissolved in 104 µL 100% DMSO (Fisher Bioreagents BP231) for a final concentration of 7.7 mM. Protein and dye were mixed in labelling buffer (20 mM HEPES pH 7.0, 200 mM KCl, 1 mM TCEP), with final concentration of p40^{phox} PX domain at 112 µM and AF647 at 224 µM and incubated for 2 hours at room temperature with slow rotation. Reaction was quenched with 1 mM DTT and loaded onto a 5 mL HiTrap Heparin HP column (Cytiva 17-0406-01) equilibrated with 20 mM HEPES pH 7.4. The column was washed with 65 mL 20 mM HEPES pH 7.4 buffer and eluted with a gradient of Hep-B buffer (20 mM HEPES pH 7.4, 1M KCl). Peak fractions were pooled and concentrated to 168 µM with 90% labelling efficiency.

Human RAB1A purification and labelling

For maleimide labelling of human RAB1A, the pJB78 plasmid (GST-TEV-HsRAB1A Q70L C26S C126S,1-204) (Tremel *et al.*, 2021 [DOI](#)) was used. To purify the protein, pJB78 was transformed into *E.coli* C41(DE3)RIPL cells, 2L of transformed bacteria were cultured in 2xTY medium supplemented 0.1 mg/mL ampicillin at 37 °C until OD₆₀₀=0.6, then induced with 0.2 mM IPTG and grown for further 18 hours at 18 °C. Cells were pelleted by centrifugation at 6700 g for 25 min, and pellets resuspended in 50 mL lysis buffer (25 mM HEPES pH 8.0, 200 mM NaCl, 1 mM TCEP, 0.5 mg/mL lysozyme (Sigma L6876), 0.05 µL/mL universal nuclease (Pierce 88702)). Cells were sonicated for 6 minutes (10 s on, 10 s off, 60% amplitude), and lysate cleared by centrifugation at 104,000 g for 45 minutes at 4 °C in a Type 45 Ti rotor (Beckmann Coulter). After filtering through a 0.45 µm syringe filter, lysate was incubated with 4 mL GS4B-GST beads equilibrated with lysis buffer for 1 hour at 4 °C at 8 rpm rotation. Beads were transferred to a gravity flow column and washed with 100 mL lysis buffer, then 100 mL wash buffer (20 mM HEPES pH 8.0, 300 mM NaCl, 5% glycerol, 1 mM TCEP). Beads were resuspended in 10mL TEV buffer (25 mM HEPES pH 8.0, 150 mM NaCl, 1 mM TCEP), 60 µL of TEV protease at 4.4 mg/mL was added, and sample was incubated overnight at 4 °C without rotation. Flow-through and five 5 mL TEV buffer elutions were collected and diluted with Hep-A buffer (25 mM HEPES pH 8.0, 1 mM TCEP) to a final NaCl concentration of 50-100 mM, then applied to a 5mL HiTrap Heparin HP column. The flow-through fraction with protein was concentrated and incubated with 16x molar excess of GTP (Jena Bioscience NU-1012) or GDP (Sigma G7127) and 27x molar excess EDTA for 90 minutes on ice, followed by addition of 55x molar excess MgCl₂ and further incubation for 30 minutes on ice. Protein was applied onto an S75 16/60 gel filtration column equilibrated with 25 mM HEPES pH 7.0, 150 mM NaCl. Main peak was pooled and concentrated to 816 µM.

To label RAB1A for affinity binding assays, 1 mg of AlexaFluor 488 C5 maleimide (Invitrogen A10254) was dissolved in 138.8 µL 100% DMSO (Fisher Bioreagents BP231) for a final concentration of 10 mM. Protein and dye were mixed in labelling buffer (20 mM HEPES pH 7.0, 200 mM KCl, 1 mM TCEP), with final concentration of RAB1A at 112 µM and AF488 at 224 µM and incubated for 2 hours at room temperature with slow rotation. Reaction was quenched with 1 mM DTT and applied to an S75 16/60 gel filtration column equilibrated with 20 mM HEPES pH 7.0, 300 mM NaCl. Peak fractions were pooled and concentrated to 88 µM with 14% labelling efficiency.

Human RAB5A purification

For maleimide labelling of human RAB5A, the pOP823 plasmid (HsRAB5A Q79L C19S,C63S, 1-212) (Tremel *et al.*, 2021 [DOI](#)) was used. To purify the protein, pOP823 was transformed into *E.coli* OverExpress™ C41(DE3) cells, 4L of transformed bacteria were cultured in 2xTY medium supplemented with 0.1 mg/mL ampicillin at 37 °C until OD₆₀₀=0.6, then induced with 0.5 mM IPTG and grown for further 20 hours at 16 °C. Cells were pelleted by centrifugation at 6700 g for 25 min, and pellets were resuspended in 50 mL lysis buffer (25 mM HEPES pH 8.0, 200 mM NaCl, 10 mM imidazole, 1 mM TCEP, 0.5 mg/mL lysozyme, 0.05 µL/mL universal nuclease). Cells were sonicated for 7 minutes (10 s on, 10 s off, 60% amplitude), and lysate was cleared by centrifugation at 104,000 g for 45 minutes at 4 °C in a Type 45 Ti rotor (Beckmann Coulter). Supernatant was applied to a 5 mL HisTrap FF column (Cytiva 17-5255-01) equilibrated with Ni-A buffer (20 mM HEPES pH 8.0, 100 mM NaCl, 5% glycerol, 10 mM imidazole, 2 mM beta-mercaptoethanol (βME)). Protein was washed with 150 mL Ni-A buffer and eluted with a gradient of Ni-B buffer (20 mM HEPES pH 8.0, 100 mM NaCl, 5% glycerol, 200 mM imidazole, 2 mM BME). Eluted sample was dialysed in presence of ULP1 SUMO protease in dialysis buffer (20 mM HEPES pH 7.0, 150 mM NaCl, 5% glycerol, 1 mM TCEP) overnight at 4 °C in 10k MWCO dialysis tubing (Thermo Fisher 68100). Dialysed protein was applied again on a 5 mL HisTrap FF column to remove the cleaved His₆-SUMO tag. Flow-through and seven 10 mL elutions with dialysis buffer were collected and concentrated using a 10k MWCO concentrator. RAB5A was loaded with GTP or GDP, purified, and concentrated in the same way as RAB1A.

To label RAB5A for affinity binding assays, 1 mg of AlexaFluor 488 C5 maleimide (Invitrogen A10254) was dissolved in 138.8 μ L 100% DMSO (Fisher Bioreagents BP231) for a final concentration of 10 mM. Protein and dye were mixed in labelling buffer (20 mM HEPES pH 7.0, 200 mM KCl, 1 mM TCEP), with final concentration of RAB1A at 112 μ M and AF488 at 224 μ M and incubated for 2 hours at room temperature with slow rotation. Reaction was quenched with 1 mM DTT and applied to an S75 16/60 gel filtration column equilibrated with 20 mM HEPES pH 7.0, 300 mM NaCl. Peak fractions were pooled and concentrated to 88 μ M with 14% labelling efficiency. Final RAB5A-AF488 product was concentrated to 146 μ M with 43% labelling efficiency.

Purification of mCherry-tagged yeast VPS34-CII

Plasmids carrying genes encoding the subunits of the WT ScVPS34-CII (pYO359 + pYO1922) or ScVPS34-CII carrying a RAB5 binding mutant (pYO1920 + pYO1922) were transformed into the YOY193 yeast strain [*ADE2::TRX1 Bcy123 (pep4::HIS3 prb1::LEU2 bar1::HISG lys2::Gal1/10-Gal4 can1 ade2 trp1 his3 ura3 leu2-3,112 lys2)*; (Rostislavleva *et al.*, 2015 [DOI](#))]. Cells were grown in YM4 medium (0.67% yeast nitrogen base without amino acids; 0.5% casamino acids and 2% glucose, supplemented with 0.002% adenine sulfate; and 0.002% tyrosine). Gene expression was induced by 2% galactose in the presence of 2% glycerol, 3% lactic acid, and 10 mM ZnCl_2 without glucose at 30 °C for 26 hours. Cells were lysed in lysis buffer [50 mM Tris at pH 8.8, 150 mM NaCl, 0.5 mM TCEP, 1% Triton-X, 0.5 mM phenylmethylsulfonyl fluoride (PMSF), and a protease inhibitor tablet (Roche 05056489001)] with zirconium beads using a cell disruptor (FastPrep-24, MP Biomedicals) at the intensity of 6.5 for 40 s. Lysates were spun at 3000 *g* for 1 min. Supernatants were transferred to 2 ml tubes, and spun at 21,000 *g* for 5 min at 4 °C on a bench top centrifuge. Supernatants were combined and passed through a 5 mm syringe filter. Immunoglobulin G (IgG) beads (GE Healthcare 17096902) were added to the lysate, incubated for 3 hours at 4 °C at 8 rpm, then washed once with wash buffer (50 mM Tris at pH 8.0, 150 mM NaCl, 0.5 mM TCEP, 0.1% Triton-X, 50 mM MgCl_2 , 5 mM ATP and 5 μ g/mL RNaseA [Sigma, 83834]) and TEV buffer (50 mM Tris, pH 8.0, 150 mM NaCl, 0.5 mM TCEP). The protein A (ZZ) tags were cleaved by TEV protease in TEV buffer overnight. The elution fractions were concentrated and subjected to gel filtration on Superdex 200 10/30 equilibrated in 20 mM HEPES at pH 8.0, 300 mM NaCl, and 0.5 mM TCEP. The peak fractions were combined and concentrated using a 100 MWCO concentrator (Millipore UFC810024).

Purification of Vps21 Q66L-EGFP

The pYO1919 plasmid was transformed into OverExpress C41(DE3) chemically competent cells (Lucigen 60442). Transformed bacteria were cultured in 4 L 2xTY medium supplemented with 0.1 mg/mL ampicillin (Formedium AMP50) at 37 °C until OD₆₀₀=0.8, then expression was induced with 0.5 mM IPTG, and bacteria were incubated for further 16 hours at 15 °C. Cells were pelleted by centrifugation at 6700 *g* for 15 minutes, frozen in liquid nitrogen, and stored at -80 °C until use. Pellets were resuspended in 50 mL lysis buffer (50 mM HEPES pH 8.0, 150 mM NaCl, 12% glycerol, 5 mM beta-mercaptoethanol (BME), 1 mM PMSF, 1x EDTA-free protease inhibitor cocktail tablet). Cells were sonicated for 5 min on ice (2 s on, 3 s off, 60% amplitude), and lysate was cleared by centrifugation at 142,000 *g* for 15 min at 4 °C in a Type 45 Ti rotor. Supernatant was filtered through a 5 μ m syringe filter then passed through a 5 ml Ni-NTA FF column (GE Healthcare). The column was washed with 50 mL wash buffer (50 mM Tris pH 8.0, 300 mM NaCl, 5 mM BME, and 12% glycerol), then the protein was eluted with 20 mL elution buffer (50 mM HEPES pH 8.5, 20 mM NaCl, 300 mM imidazole, and 5 mM BME). The eluted protein was immediately loaded on a 5 mL Q column (GE Healthcare). The bound protein was eluted with a NaCl gradient between Q-A buffer (50 mM Tris pH 8.5 and 5 mM BME) and Q-B buffer (50 mM Tris pH 8.5, 1 M NaCl, and 5 mM BME). Elution fractions were combined and concentrated down to 219 mg/ml (4.3 mM). The protein was diluted to 1 mM, and incubated with 11 mM GTP and 18 mM EDTA at rt for 90 min, then 36 mM MgCl_2 was added and further incubated for 30 min at rt. The GTP-loaded protein was subjected to gel filtration on a S75 16/60 column in GF buffer (20 mM HEPES 8.0, 150 mM NaCl, and 0.5 mM TCEP). After combining the peak fractions, 10 mM TCEP at a final concentration was added and the protein was concentrated down to 130 mg/ml (2.57 mM).

Multiple sequence alignment

Sequences for human RAB GTPases were retrieved from UniProtKB (<https://www.uniprot.org/>; (UniProt, 2025)) and multiple sequence alignments generated using Clustal Omega (<https://www.ebi.ac.uk/jdispatcher/msa/clustalo> (Sievers & Higgins, 2021)). Alignments were inspected and corrected using JalView (Waterhouse *et al.*, 2009), and results were analysed and visualised with ESPript 3.0 (<https://esprict.ibcp.fr/ESPript/ESPript/>; (Gouet *et al.*, 1999)).

GUV assay

GUVs were prepared using a hydrogel-assisted swelling protocol. 88 μ L of 5% polyvinyl alcohol (PVA, Sigma 8.14894, MW approx. 145 kDa) solution was deposited onto a 13 mm cleaned coverslip (VWR 631-0150) and a thin layer of PVA was created by centrifuging for 30 s in a fixed-speed microfuge (IKA mini G, 6000 rpm) using a custom-made adaptor for cover slips. PVA-coated coverslips were dried at 60 °C for 30 minutes, and 15 μ L of GUV lipid mixture at 1 mg/mL lipid concentration was deposited on the PVA layer. Coverslips were desiccated for 60 minutes, and GUVs swollen for 1 hour at room temperature in 220 μ L 500 mM sterilised sucrose solution. Freshly generated GUVs were transferred to a 1.5 mL tube passivated by incubating with 1 mL of 5 mg/mL BSA (Sigma A7030) for 1 hour, followed by two 500 mM sucrose rinses.

To immobilise the GUVs, chambers of an 8-well glass bottom plate (Ibidi 80807) were incubated with 150 μ L of avidin solution (0.1 mg/mL egg-white avidin, Invitrogen A2667, dissolved in PBS, and 1 mg/mL BSA) for 15 minutes at room temperature, and washed three times with 150 μ L observation buffer (50 mM HEPES pH 7.0, 241.4 mM NaCl). 64 μ L observation buffer was mixed with 48 μ L GUVs in each chamber and incubated overnight at 4 °C in presence of 10 μ M (4-fold molar excess to the PE-MCC lipid) RAB protein. GUVs were carefully washed four times with 360 μ L wash buffer (31.8 mM HEPES pH 8.0, 172.7 mM NaCl, 181.8 mM sucrose, 5 mM BME).

GUVs were observed with a 63x oil-immersion objective (Plan-Apochromat 63x/1.4 Oil DIC M27, Zeiss) on an inverted confocal microscope (LSM 880 Indimo, AxioObserver, Zeiss), using ZEN software (Zeiss). The observation chamber was immobilised on a universal mounting frame (PeCon 010-800 486) using a chamber system insert plate (PeCon 000 470). After confirming that GUVs were immobilised on the wells, 20 μ L of 10x buffer was added (250 mM HEPES pH 8.0, 20 mM MnCl_2 , 10 mM TCEP, 1 mM ATP pH 8.0, 10mM EGTA in assays with EGTA). In ZEN software, Time Series and Positions were selected, and Definite Focus.2 (Zeiss) was used to maintain focus for every time point and position. The Lissamine-Rhodamine channel for GUV visualisation was excited using a diode-pumped solid-state (DPSS) 561 nm laser and collected with a 570-623 nm band. The AlexaFluor 647 channel for PX domain detection was excited using a HeNe 633 nm laser and collected with a 638-755 nm band. Four areas were selected per well to analyse at least 10 GUVs. To start the reaction, 48 μ L of a solution containing the VPS34 kinase complex, labelled PX domain, in protein dilution buffer (25 mM HEPES pH 8.0, 150 mM NaCl, 1 mM TCEP, 0.67 mg/mL BSA) was added to the chamber. No BSA was added in the protein dilution buffer for assays without BSA. The final concentration of PX domain was 3.7 μ M. The final concentrations of VPS34 complexes are described in figures and their legends. Images of each position were taken every 120 s for 2-3 h. Images were analysed using the strategy and macro described in (Ohashi *et al.*, 2020).

Confocal microscopy bead-based assay for RAB protein binding to VPS34 complexes

10-30 μ L RFP-trap agarose beads (Chromotek RTA-10) were washed with 100 μ L binding buffer (20 mM HEPES pH 8.0, 100 mM NaCl, 0.5 mM TCEP) and incubated with 2 μ M mCherry-tagged VPS34 complex for 30 min on ice. Excess buffer was removed after centrifugation at 500g for 30 s and beads were washed with 100 μ L binding buffer to remove unbound VPS34 complex. 30 μ L reactions were prepared in PCR tubes with final bead concentration of 2% and AF488-labelled RAB proteins were added at varying final concentrations. Reactions were incubated on ice for 30 min and transferred to a 384-well plate (Greiner 781856) for imaging.

Beads were imaged with a 10x objective (EC Plan-Neofluar 10x/0.3 M27, Zeiss) on an inverted confocal microscope (LSM 880 Indimo, AxioObserver, Zeiss), using ZEN software (Zeiss). The 384-well plate was immobilised on a universal mounting frame (PeCon 010-800 486). DIC channel was used to visualise beads. The mCherry channel was excited using a diode-pumped solid-state (DPSS) 561 nm laser and collected with a 578-696 nm band. The AlexaFluor 488 channel was excited using an Argon 488 nm laser and collected with a 493-543 nm band. Images were analysed using Fiji software. Beads were selected using the ellipse ROI (region of interest) selection tool and added to the ROI manager, and fluorescence intensity of mCherry and AF488 channels was measured using a modified version of macro described by Ohashi et. al. (Ohashi *et al.*, 2020 [DOI](#)). Background intensity was measured from 10 ROIs without beads and averaged to subtract from measured bead fluorescence intensities. For each bead, intensity of the AF488 channel was normalised by mCherry channel intensity to account for unequal mCherry-tagged VPS34 complex loading on the RFP-trap beads. Data were further analysed with GraphPad Prism (GraphPad) and binding affinities estimated using the non-linear fitting tool with 'One site – specific binding' option.

Nano differential scanning fluorimetry

To determine the melting temperatures (T_m) of VPS34 complexes, nanoDSF experiments were carried out on the Prometheus NT.48 instrument (NanoTemper). 11 μ L samples were prepared with final VPS34 complex concentration of 0.5 mg/mL in SEC buffer (20 mM HEPES pH 8.0, 0.5 mM TCEP, 150 mM NaCl for complex I or 300 mM NaCl for complex II) and incubated on ice for 30 min. Prometheus standard capillaries (NanoTemper PR-C002) were filled with 10 μ L sample and placed on the sample holder. Prior to the experiment, fluorescence signal was optimised by adjusting the excitation power. A temperature gradient was then applied at 1°C/min rate, from 15°C to 90°C, and intrinsic protein fluorescence emissions recorded at 330 nm and 350 nm. Data were visualised and analysed in GraphPad Prism (GraphPad).

Yeast strains

The *S. cerevisiae* BY4741 (MATa *his3 Δ 1 leu2 Δ 0 met15 Δ 0 ura3 Δ 0*) strain was from Open Biosystems. The *vps15 Δ* null yeast was created by a PCR-based gene targeting with the primers yoo1652 (5'-GTT GAATTGGAGAAATACAAAAGCTGTAAGGTTATCAAAAAGGAAGGCATACAGT ATAgtttagcttgctgctcccgcc -3') and yoo1653 (5'-GAATACAAATTATACGCATTTAGAATAAAGAAGCTGaaactggatggcggcgtagtat-3'), and the *Schizosaccharomyces pombe* *HIS5* gene as a template (Longtine *et al.*, 1998 [DOI](#)). For fluorescence microscopy, a *vps15 Δ* strain integrated with a plasmid carrying N-terminally tandem-tomato-tagged VPS21 Q66L (pYO1879) was created by linearising the plasmid with NheI enzyme (YOY415).

Yeast plasmid transformation

For each transformation, 5 - 10 absorbance units at 600 nm (A_{600}) of yeast cells either from plates or liquid culture were suspended in 100 μ L One-step buffer (40% PEG3350, 0.2 M lithium acetate, 10 mM Tris pH 7.4, 100 mM DTT), and 3 μ L of a plasmid was added to the mixture and incubated at room temperature for 10 min. The mixture was then incubated at 42 °C for 30 min, spun at 3000 g for 30 s to remove the buffer. The yeast pellet was resuspended in 50 mL H₂O and spread on a -URA plate.

CPY assay

Yeast cells were grown to log phase (A_{600} = 0.5-0.9) at 30 °C in -URA medium. Five A_{600} units of cells were collected and resuspended in 500 μ L of -URA medium containing 0.5 mg/mL bovine serum albumin (BSA) and 50 mM KPO₄ at pH 5.7 and incubated at 30 °C for 1 hour. Cell and medium fractions were separated by centrifugation at 3000g for 1 min. Medium fractions were precipitated with final 10% trichloroacetic acid (TCA, from 100% stock) on ice for 15 min and centrifuged at 4 °C at 20,000g for 10 min. Pellets were washed once with 90% cold acetone. Both fractions were resuspended in 100 μ L of lysis buffer (50 mM Tris at pH 8.0, 1% SDS, 8 M urea, 10 mM DTT, and 5

mM EDTA), 30 μ L of 4x sample buffer, and 0.5 mm zirconia beads (YTZ® GRINDING MEDIA, Tosh). Cells and TCA pellets were disrupted at 5000 revolutions per minute (rpm) for 30 s with PowerLyzer (Mo Bio) at 4°C. Samples were boiled at 98 °C for 2 min and loaded on SDS-PAGE gel.

CPY is synthesized as a premature form (p1), glycosylated in the Golgi (p2), and then delivered to the vacuole, where it is cleaved to become a mature form (m). The m and p2 forms of CPY were detected both in cellular and medium fractions using anti-CPY (1/1000; Invitrogen A6428). The C-terminally FLAG-tagged Vps15 proteins were detected by anti-Flag-HRP (HRP, horseradish peroxidase) (1/1000; Sigma A8592). Pgk1 was detected by anti-Pgk1 antibody (1/1000; Abcam ab113687). The bands were detected using ChemiDoc™ Touch Imaging System (Bio-Rad) with Chemiluminescence mode.

For each condition, intensities of p2 and m bands in cell and medium fractions were summed as a total intensity, then the percentages of m and p2 were calculated. In the case p2 was detected both in cell and medium fractions, the percentage of the summed intensity from these fractions was shown. The band intensities were measured using Image Lab 6.0 (Bio-Rad). The quantification was from 3 independent experiments.

Yeast fluorescence microscopy

The YOY415 strain was transformed with the plasmids carrying WT VPS15-EGFP (pYO634) or RAB5 binding mutant VPS15-EGFP (pYO1880). Cells were grown to mid-log phase ($A_{600}=0.5-0.9$), then examined with an inverted microscope (Nikon Eclipse Ti2) equipped with a CMOS camera (HAMAMATSU ORCA-Flash4.0 C11440) and a niji LED Light Source (bluebox Optics). Images were analysed and levels were adjusted using Fiji.

Hydrogen-deuterium exchange coupled to mass spectrometry

HDX-MS experiments were carried out as follows: Complex II (48 μ M) alone or in the presence of 90 μ M RAB5A-Q79L-C19S-C63S-GTP were pre-incubated for 30 min on ice. Complex II alone sample was then exposed for different deuteration times (3 sec on ice, 3, 30, 300 and 3000 sec at room temperature) to D₂O only buffer (20 mM Hepes pD8, 300 mM NaCl) to yield solution that was 5 μ M Complex II and D₂O at 83% final concentration. Complex II + RAB5A sample was exposed for the same deuteration times to a D₂O + RAB5A buffer (20 mM Hepes pD8, 300 mM NaCl, 90 μ M RAB5A) to yield solution that was and 5 μ M Complex II + 90 μ M RAB5A and D₂O at 83% final concentration. Samples were then immediately flash-frozen in liquid nitrogen and stored at -80°C until analysis. Prior to LC-MS analysis, samples were quickly thawed and injected on a HDX Manager coupled to an Acquity UPLC M-Class system (Waters) set at 0.1 °C. Samples were then digested online at 15 °C using a 20 mm x 2.0 mm home-made pepsin column (Pepsin from Thermo Scientific, 20343 and hardware from Upchurch Scientific, C130-B) and loaded on a UPLC pre-column (ACQUITY UPLC BEH C18 VanGuard pre-column, 2.1 mm I.D. x 5 mm, 1.7 μ m particle diameter, Waters, 186003975) for 2 min at 100 μ L/min. Digested peptides were then eluted over 20 min from the pre-column onto an ACQUITY UPLC BEH C18 column (1.0 mm I.D. x 100 mm, 1.7 μ m particle diameter, Waters, 186002346) using a 5-43% gradient of acetonitrile (Romil, H050), 0.1% formic acid. MS data were acquired using a Synapt G2Si-HDMS (Waters) with electrospray ionization, using the data-independent acquisition mode (HDMS^F) over an m/z range of 50-2000 and a 100 fmol/ μ L Glu-FibrinoPeptide solution was used for lock-mass correction and calibration. The following parameters were used during the acquisition: capillary voltage, 3 kV; sampling cone voltage, 40 V; source temperature, 90 °C; desolvation gas, 150 °C and 650 L.h⁻¹; scan time, 0.3 s; trap collision energy ramp, 20–45 eV. Peptide identification was performed using ProteinLynx Global Server 3.0.1 (PLGS, Waters) with a home-made protein sequence library containing VPS34, VPS15, UVRAG, BECLIN1, RAB5A-Q79L-C19S-C63S-GTP, and pepsin sequences, with peptide and fragment tolerances set automatically by PLGS. Peptides were filtered using HDEaminer 3.4.0 (Trajan Scientific and Medical) with a minimum fragment per amino acid of 0.2, a minimum intensity of 10³, a length between 5 and 25 residues, a minimum PLGS score of 6.62, and an MH⁺ tolerance of 5 ppm. An initial automated spectral processing step was conducted by HDEaminer followed by a manual inspection of individual peptides for sufficient quality where only one charge state per

peptide was kept. Deuterium uptakes were not corrected for back-exchange and are reported as relative. HDX-MS results were statistically validated using an in-house program (Archaeopteryx), where significant differences required 0.3 Da, 5% and p-value of <0.05 (n=3). A table with peptides within the dataset and the quality assessment statistics are available in Supplementary Data 1. HDX-MS results were illustrated on a Complex II model using PyMOL 2.5.4 (www.pymol.org). The mass spectrometry proteomics data (including raw data, table of every peptide included within the dataset and all relative uptake plots) have been deposited to the ProteomeXchange Consortium via the PRIDE partner repository (Perez-Riverol *et al.*, 2022) with the dataset identifier PXD061277.

Cryo-ET and subtomogram averaging

We re-processed the cryo-ET dataset of Complex II-BATS on RAB5-coated LUVs described in (Tremel *et al.*, 2021). The tilt series raw images were imported in Relion 5.0 (beta4-commit-dfea29, (Burt *et al.*, 2024)) --tomo mode, and motion corrected using Relion 5.0 implementation of MOTIONCOR2 (Zheng *et al.*, 2017). CTF was estimated using the build-in CTFFIND-4.1 (Rohou & Grigorieff, 2015), and tilt series were aligned using the IMOD (Kremer *et al.*, 1996) wrapper with fiducial based alignment. 10 Å/pixel tomograms were reconstructed for picking, and particles on membranes were manually picked on 3 representative tomograms. CrYOLO 1.9.9 (Wagner *et al.*, 2019) was used for picking in tomograms after training on the manually picked particles using the trained custom model described above as pretrained weights, with a threshold of 0.05 and a box size of 220 Å. 146,464 particles were re-imported in Relion 5.0, scaled to the raw movies and extracted at bin 8 with a box size of 375 Å. The EMD-12237 density was used as a reference for 3D classification using a low-pass filter of 60 Å, the best 3D class was selected with 23,237 particles and subjected to another round of 3D classification using bin 4 particles. Two classes with 8,441 particles were selected and refined at bin 1, masking out the membrane and using Blush regularisation. One refinement round of CTF refinement followed by Bayesian polishing was done and reached a resolution of 9.88 Å. To look for the Rab5 binding site in VPS15, a 3D classification was done on the 23,237 bin4 particles using the SPA structure of Complex II with two RAB5 molecules as a 40 Å low passed template. To search for alternative conformations of Complex II on the membrane, a 3D classification was done on the same particles using the membrane bilayer as a reference.

Cryo-EM sample preparation and screening

Complex II VPS34 199-REIE-202 > 199-AAAA-202 + RAB5A and Complex II VPS34 199-REIE-202 > 199-ERIR-202 + RAB5A

VPS34-CII at the final concentration of 6 µM was incubated with 90 µM RAB5A for 20 min on ice in buffer containing 50 mM HEPES pH 8.0, 200 mM NaCl, 1 mM TCEP, 5 mM MgCl₂, and 1 mM AMP-PNP. The sample was crosslinked with BS3 at a final concentration of 30 µM for further 20 min on ice, then quenched with 100 mM TRIS pH 8.0. Prior to blotting, the sample was supplemented with 0.005% (w/w) Nonidet P-40 substitute and 4 mM CHAPSO. 3.5 µL sample was applied to a glow-discharged (40 mA, 90 s, Edwards Sputter Coater S150B) UltrAuFoil R 1.2/1.3 Au300 grid and plunge-frozen in liquid ethane using a Vitrobot Mk 3 (Thermo Fisher) plunger at 14 °C, 100% humidity, with 0 s waiting time, 4 s blotting time, and blot force set to +8. Grids were screened on a 200 kV Glacios (Thermo Fisher) microscope with a Falcon 3 detector in linear mode.

WT Complex II + RAB5A

VPS34-CII at the final concentration of 4 µM was incubated with 90 µM RAB5A for 20 min on ice in buffer containing 50 mM HEPES pH 8.0, 200 mM NaCl, 1 mM TCEP, 5 mM MgCl₂, and 1 mM AMP-PNP. The sample was crosslinked with BS3 at a final concentration of 20 µM for further 20 min on ice, then quenched with 100 mM TRIS pH 8.0. Prior to blotting, the sample was supplemented with 0.005% (w/w) Nonidet P-40 substitute and 4 mM CHAPSO. 3.5 µL sample was applied to a glow-discharged (40 mA, 90 s, Edwards Sputter Coater S150B) UltrAuFoil R 1.2/1.3 Au300 grid and

plunge-frozen in liquid ethane using a Vitrobot Mk 3 (Thermo Fisher) plunger at 14 °C, 100% humidity, with 0 s waiting time, 4 s blotting time, and blot force set to +8. Grids were screened on a 200 kV Glacios (Thermo Fisher) microscope with a Falcon 3 detector in linear mode.

Cryo-EM data collection

Complex II VPS34 199-REIE-202 > 199-AAAA-202 + RAB5A

Images were collected on a 300 kV Titan Krios G3 TEM (Thermo Fisher) with a Gatan K3 detector (Gatan) in counting mode and energy filter set to 20 eV. 10,739 movies were collected at 105,000x magnification and pixel size of 0.86 Å / px using EPU software (ThermoFisher) in AFIS mode, with 2 movies collected per foil hole. The defocus range used was -0.8 nm to -2.2 nm, and the dose rate was 15.1 e⁻ / px / s over an exposure length of 1.96 s, making the total dose 40 e⁻ / Å. Movies were fractionated into 40 frames.

Complex II VPS34 199-REIE-202 > 199-ERIR-202 + RAB5A

Images were collected on a 300 kV Titan Krios G3 TEM (Thermo Fisher) with a Gatan K3 detector (Gatan) in counting mode and energy filter set to 20 eV. 13,301 movies were collected at 105,000x magnification and pixel size of 0.86 Å / px using EPU software (ThermoFisher) in AFIS mode, with 3 movies collected per foil hole. The defocus range used was -0.8 nm to -2.2 nm, and the dose rate was 26.4 e⁻ / px / s over an exposure length of 1.12 s, making the total dose 40 e⁻ / Å. Movies were fractionated into 40 frames.

WT Complex II + RAB5A

Images were collected on a 300 kV Titan Krios G1 TEM (Thermo Fisher) with a Gatan K3 detector (Gatan) in counting mode and energy filter set to 20 eV. In total, 19,844 movies were collected on two grids, at 105,000x magnification and pixel size of 0.73 Å / px using EPU software (ThermoFisher) in AFIS mode, with 3 movies collected per foil hole. The defocus range used was -0.8 nm to -2.2 nm. For the first grid, the dose rate was 15 e⁻ / px / s over an exposure length of 1.42 s, making the total dose 40 e⁻ / Å. For the second grid, the dose rate was 21.5 e⁻ / px / s over an exposure length of 1 s, making the total dose 40 e⁻ / Å. Movies were fractionated into 40 frames.

Cryo-EM data processing

VPS34-CII VPS34 199-REIE-202 > 199-AAAA-202 + RAB5A (Fig. S10A [↗](#))

Data was pre-processed using RELION 4.0. Movie frames were motion-corrected using MotionCorr2, and CTF of motion-corrected micrographs was estimated using CTFFIND-4.1. Micrographs with CTF-estimated maximum resolution of ≤5 Å were selected for further processing. Particle picking was done with CrYOLO-1.7.5 using a custom-trained model. Particles were extracted with two-fold downsampling and at a box size of 204 x 204 px and imported to CryoSPARC for 2D classification. Selected particles were converted from .cs to star and re-imported to RELION for re-extraction at full box size of 408 x 408 px. Full size particles were again imported into CryoSPARC and used for homogeneous refinement with the density of VPS34-CI low-passed to 40 Å as an initial reference, yielding a 3.43 Å map after FSC-mask auto-tightening. An initial model for complex II REIE > AAAA + RAB5A was built into this density by rigid-body fitting PDB depositions of VPS34 kinase domain (PDB: 3IHY) and RAB5A (PDB: 1R2Q), AlphaFold2 predictions of VPS15, VPS34 C2 domain, BECLIN1 CC2-BARA domains, UVRAG CC2-BARA2-CTD domains, and AlphaFold3 prediction of BECLIN1 CC1 domain and UVRAG-CC1 domain. Models were consolidated into a single entity for complex II REIE > AAAA + RAB5A using Coot 0.9.8.7. From this initial model, UCSF ChimeraX command ‘molmap’ was used to generate masks on the adaptor arm, kinase arm, base of the complex, and the RAB5A binding site on VPS34. The particles from the homogeneous refinement job in CryoSPARC were classified into eight 3D classes, and selected classes (see Fig. S10A [↗](#) for details) were used to refine regions of the complex mentioned above. The adaptor arm was refined to 3.2 Å resolution; the kinase arm yielded a map with resolution of 3.15 Å; the map of the base of the complex had a resolution of 3.2 Å; the RAB5A binding site on VPS34 was refined to 3.25 Å. Maps from the consensus and local refinements were processed using CryoTEN. Local

refinement maps were merged into a composite map in UCSF ChimeraX. The initial model for complex II REIE > AAAA + RAB5A was refined using the composite map with ISOLDE, StarMap, TEMPy-ReFF, PHENIX, and Servalcat. Model was validated through MolProbity.

Two of the eight classes (classes 1 and 6) from the final 3D classification job were later found to contain RAB5A bound to VPS15 subunit. These densities were used as initial references for heterogeneous refinement in the processing of complex II REIE>ERIR + RAB5A and wild type complex II + RAB5A datasets. One class did not contain RAB5A bound to the complex, and this density was used as an ‘apo complex II’ reference.

Complex II VPS34 199-REIE-202 > 199-ERIR-202 + RAB5A (Fig. S10B [↗](#))

Movie frames were motion-corrected in RELION 4.0 with MotionCorr2, and CTF of motion-corrected micrographs was estimated using CTFFIND-4.1. Micrographs with CTF-estimated maximum resolution of ≤ 5 Å were selected for further processing. Particle picking was done with CrYOLO-1.7.5 using a custom-trained model. Picked particles were extracted with two-fold downsampling and a box size of 220 x 220 px, then imported to CryoSPARC for heterogeneous refinement with five initial references: a noise decoy, apo complex II, complex II + RAB5A on VPS34, complex II + RAB5A on VPS15, and complex II with two RAB5A bound simultaneously. The particles sorted into the classes with noise decoy and complex II + (RAB5A)₂ volumes did not produce meaningful densities and were removed. To find particles that had RAB5A bound to VPS34, the other three classes were subject to 3D classification into two classes with a focus mask on the RAB5A binding site on VPS34, and ‘force hard classification’ parameter set to ‘true’. The class with a density resembling a RAB5A bound to VPS34 was poorly resolved and further processing of the particles associated with this density did not yield a meaningful map. Therefore, the class without a density for RAB5A on VPS34 was selected for further processing. These particles were again force classified into two 3D classes, with a focus mask on RAB5A bound to VPS15. Particles from the class with an extra density on the VPS15 subunit were re-imported to RELION and re-extracted at full size with box size of 440 x 440 px, then imported back to CryoSPARC for non-uniform refinement that yielded a map at 3.68 Å after the FSC-mask auto-tightening. An initial model was built into this density by fitting the model of complex II REIE>AAAA + RAB5A. The RAB5A chain was moved from the binding site on VPS34 subunit to the VPS15 subunit and fitted into the density. This model was used in UCSF ChimeraX to generate masks for local refinements via the ‘molmap’ command. The base of the complex was refined to 3.52 Å; the kinase arm to 3.55 Å, the adaptor arm to 4.09 Å, and the local refinement of the RAB5A binding site on the VPS15 subunit yielded a map at 3.52 Å. Maps from the consensus and local refinements were processed using CryoTEN. Densities from local refinements were combined into a composite map on UCSF ChimeraX, which was used to refine the initial model for complex II REIE > ERIR + RAB5A with ISOLDE, StarMap, TEMPy-ReFF, PHENIX, and Servalcat. The model was validated through MolProbity.

WTVPS34-CII + RAB5A (Fig. S10C [↗](#))

Two datasets were collected for this sample. For both datasets, the motion-correction was done in RELION 4.0 with MotionCorr2, and the CTF of motion-corrected micrographs was estimated using CTFFIND-4.1. Micrographs with CTF-estimated maximum resolution of ≤ 5 Å were selected for further processing. Particle picking was done with CrYOLO-1.7.5 using a custom-trained model. Picked particles were extracted with two-fold downsampling and a box size of 220 x 220 px, then imported to CryoSPARC.

For dataset 1, two rounds of heterogeneous refinement were performed with five initial references: a noise decoy, apo complex II, complex II + RAB5A on VPS34, complex II + RAB5A on VPS15, and complex II with two RAB5A bound simultaneously. Least noisy classes were further classified into ten 3D classes. Particles from seven classes were re-imported into RELION and re-extracted at full size with box size of 440 x 440 px, then imported back to CryoSPARC for merging with dataset 2.

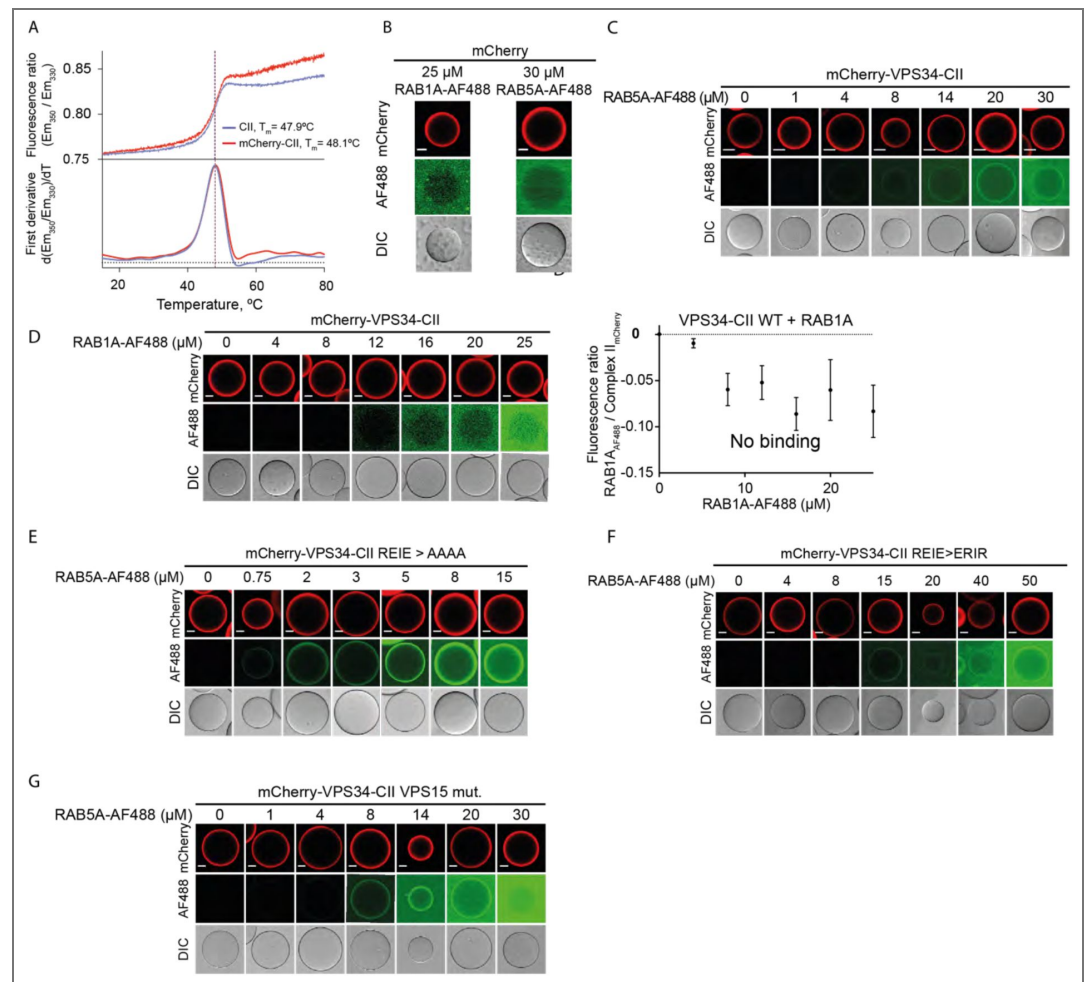


Figure S1. RAB binding by wildtype and mutant VPS34 complexes. **A.** mCherry tagging does not compromise VPS34 complex structural integrity. nanoDSF melting curves for VPS34-CII WT, $T_m=47.9^\circ\text{C}$ (blue) and mCherry-tagged VPS34-CII $T_m=48.1^\circ\text{C}$ (red). **B-G.** Micrographs of RFP-trap bead based binding assays for RAB binding to mCherry-tagged VPS34 complexes. Top row shows mCherry channel fluorescence intensity, confirming mCherry-tagged protein immobilisation on RFP-trap beads. Middle row shows AF488 channel intensity from AF488-RAB proteins. Bottom row shows the brightfield images of beads. Scale bars in all micrographs are 20 μm . **B.** mCherry protein immobilised on RFP-trap beads does not bind RAB1A or RAB5A. **C.** RAB5A binding to WT VPS34-CII. **D.** RAB1A binding to WT VPS34-CII is too weak to be quantified. Left: micrographs of the binding assay. Right: fluorescence intensity of the AF488 channel shown in micrographs on the left was normalised by the intensity of the mCherry channel and plotted against the concentration of AF488-RAB1A. **E.** RAB5A binding to VPS34-CII carrying a 199-REIE-202 > 199-AAAA-202 mutation in VPS34. **F.** RAB5A binding to VPS34-CII carrying a 199-REIE-202 > 199-ERIR-202 mutation in VPS34. **G.** RAB5A binding to VPS34-CII carrying a 579-SHMIT-583 > 579-DDMIE-583 mutation in VPS15.

For dataset 2, picked particles were subject to 2D classification, and then used in homogeneous refinement together with complex I + NRBF2 MIT domain as the initial reference. Particles were then classified into two 3D classes with a focus mask on the RAB5A binding site on VPS34. The class with a more prominent density on the VPS34 RAB5 binding site was used in two rounds of heterogeneous refinement, with four initial references: complex II + RAB5A on VPS15, complex II + RAB5A on VPS34, complex II with two RAB5A bound simultaneously, and apo complex II. Particles from the best densities were re-imported to RELION, re-extracted at full size with the box size of 440 x 440 px, imported back to CryoSPARC and merged with particles from dataset 1.

Combined 233,866 particles were subject to two rounds of heterogeneous refinement using five initial references: a noise decoy, apo complex II, complex II + RAB5A on VPS34, complex II + RAB5A on VPS15, and complex II with two RAB5A bound simultaneously. 66,323 particles were classed as

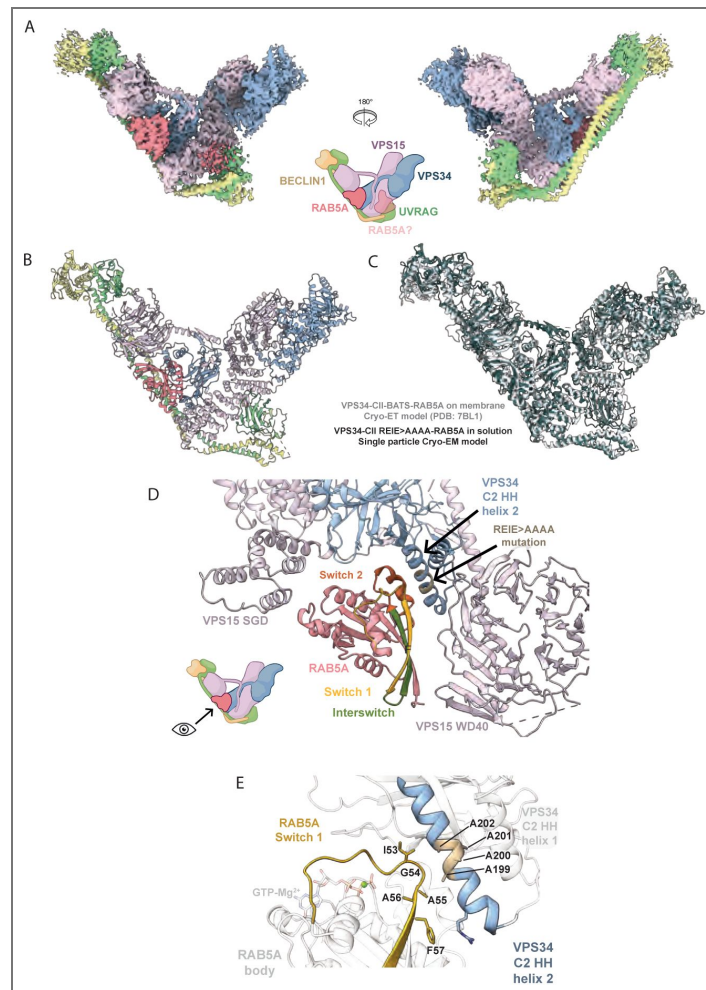


Figure S2. Analysis of the cryo-EM structure of VPS34-CII with the VPS34 199-REIE-202 > AAAA mutation bound to RAB5A.

Icons of VPS34-CII are based on cartoons from *BioRender*. **A.** A composite cryo-EM map of VPS34-CII REIE>AAAA with RAB5A bound to VPS34 C2 HH. **B.** Cartoon representation of the atomic model for VPS34-CII REIE>AAAA bound to RAB5A. Model was built using PDB depositions of VPS34 kinase domain (VPS34 residues 539-887; PDB: 3IHV) and RAB5A (PDB: 1R2Q), AlphaFold Multimer predictions of VPS15, VPS34 C2 domain (VPS34 residues 1-233), Beclin1 CC2-BARA domains (Beclin1 residues 172-269 for CC2 domain, 270-449 for BARA domain), UVRAG CC2-BARA2-CTD domains (UVRAG residues 240-333 for CC2 domain, 334-464 for BARA2 domain, and 465-699 for CTD), and AlphaFold3 prediction of Beclin1 CC1 domain (Beclin1 residues 139-171) and UVRAG CC1 domain (UVRAG residues 195-226). **C.** Overlay of single particle cryo-EM derived model of VPS34-CII REIE>AAAA-RAB5A (dark grey) and cryo-ET derived model of membrane bound VPS34-CII-BATS-RAB5A (light grey; PDB: 7BL1). Models are aligned on VPS34 helical and kinase domains (VPS34 residues 292-887). **D.** Tripartite site of RAB5A binding to VPS34-CII REIE>AAAA. RAB5A is situated close to VPS15 SGD (VPS15 residues 723-805) and WD40 domains (VPS15 residues 968-1358), and VPS34 C2 HH (VPS34 residues 170-210). RAB5A Switch 1 (yellow) and Switch 2 (orange) regions are oriented towards VPS34 C2 HH helix 2 (VPS34 residues 191-210) that carries the REIE>AAAA mutation (brown). VPS34 C2 HH helix 1 (VPS34 residues 170-184) does not contact RAB5A. **E.** Close-up of RAB5A Switch 1 (yellow) and VPS34 C2 HH helix 2 (blue) interaction, with other elements in the binding site coloured grey. RAB5A Switch 1 is rich in aliphatic and hydrophobic side chains that interact favourably with the charged to non-polar type REIE>AAAA mutation (brown) in VPS34 C2 HH helix 2.

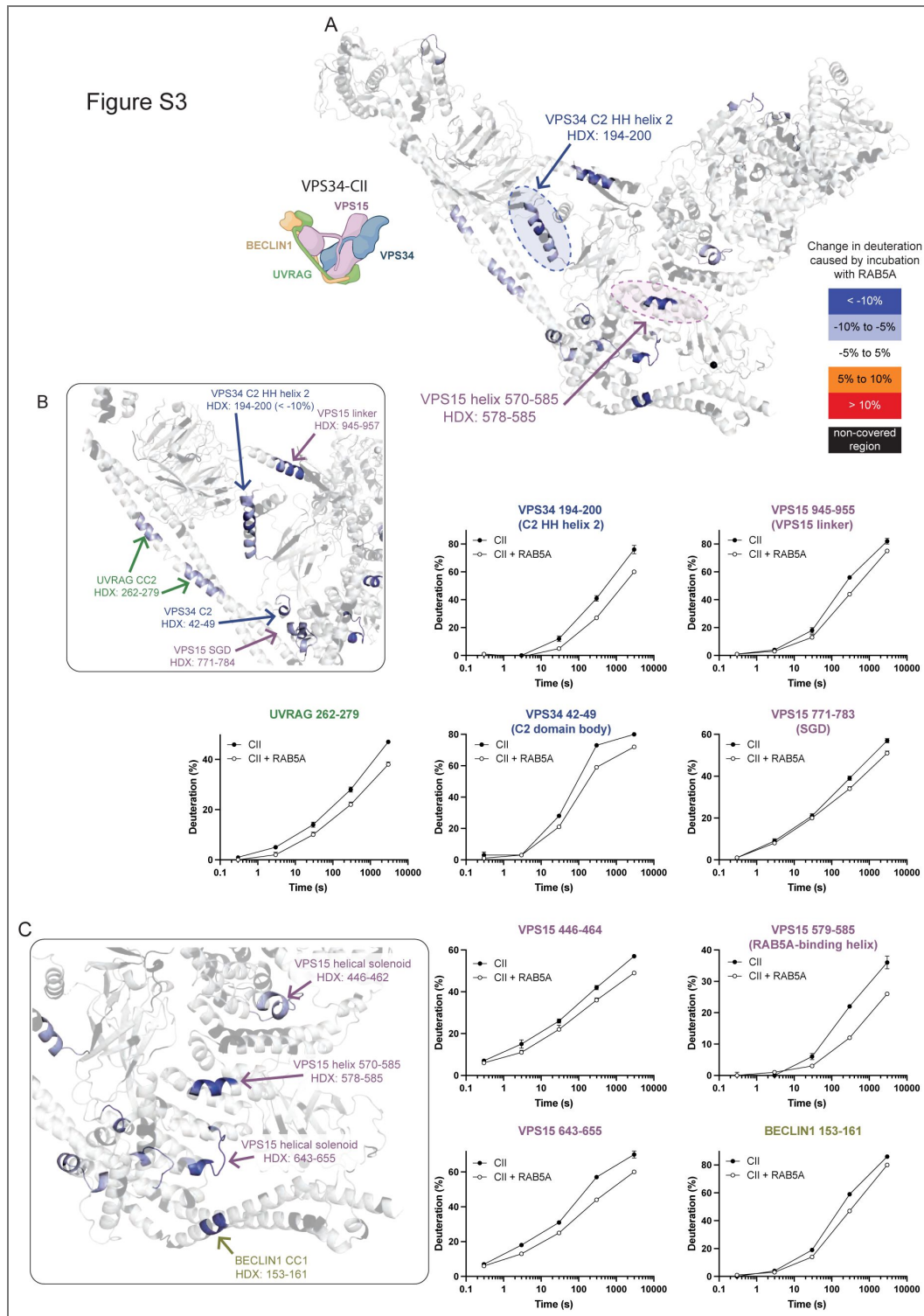


Figure S3. HDX-MS shows protection in the two RAB5A binding regions in WT VPS34-CII.

Icons of VPS34-CII are based on cartoons from *BioRender*. **A.** RAB5A binding strongly protects the two RAB5A binding sites observed in cryo-EM structure: the VPS34 C2 HH helix 2 (highlighted in blue) and VPS15 helix 570-585 (highlighted in pink). Changes in deuteration are displayed as a relative change between VPS34-CII-RAB5A and apo VPS34-CII. The changes are mapped on a model of WT VPS34-CII apo. **B.** A close-up of the VPS34 binding site for RAB5A on VPS34-CII. In addition to the VPS34 C2 HH helix 2, RAB5A binding protects other nearby elements: VPS15 SGD, VPS15 linker, UVRAG CC2, and VPS34 C2 domain. Deuterium uptake plots are shown for these regions. **C.** A close-up of the VPS15 binding site for RAB5A on VPS34-CII. RAB5A binding strongly protects VPS15 helix 570-585 and VPS15 helical solenoid residues 446-462 and 643-655. Beclin1 CC1 is also protected. Deuterium uptake plots are shown for these regions.

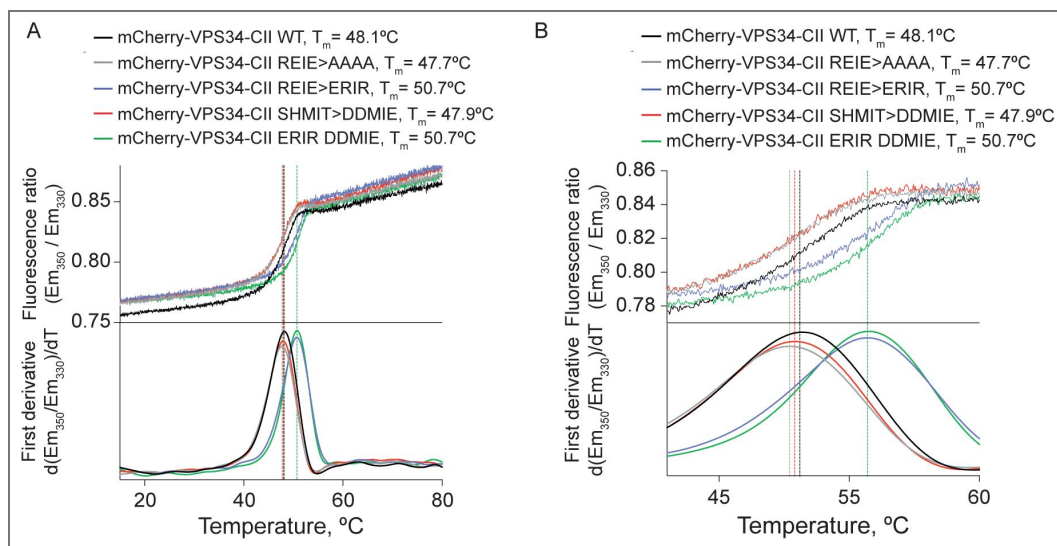


Figure S4. nanoDSF experiments show that VPS34-CII mutants that disrupt RAB5 binding are structurally intact.

A. Mutations to modify RAB5A binding to VPS34-CII either do not affect or increase the stability of VPS34-CII. The VPS34 199-REIE-202 > 199-AAAA-202 and VPS15 579-SHMIT-583 > 579-DDMIE-583 mutation in VPS34-CII did not affect the melting temperature of the complex. The VPS34 199-REIE-202 > 199-ERIR-202 mutation increased the melting temperature of VPS34-CII by 2.6°C. The 'ERIR DDMIE' label (green line) denotes the double VPS34-CII mutant carrying both VPS34 199-REIE-202 > ERIR and VPS15 579-SHMIT-583 > DDMIE mutations. **B.** Plots shown in **A** with a smaller temperature range on the x-axis for closer curve inspection.

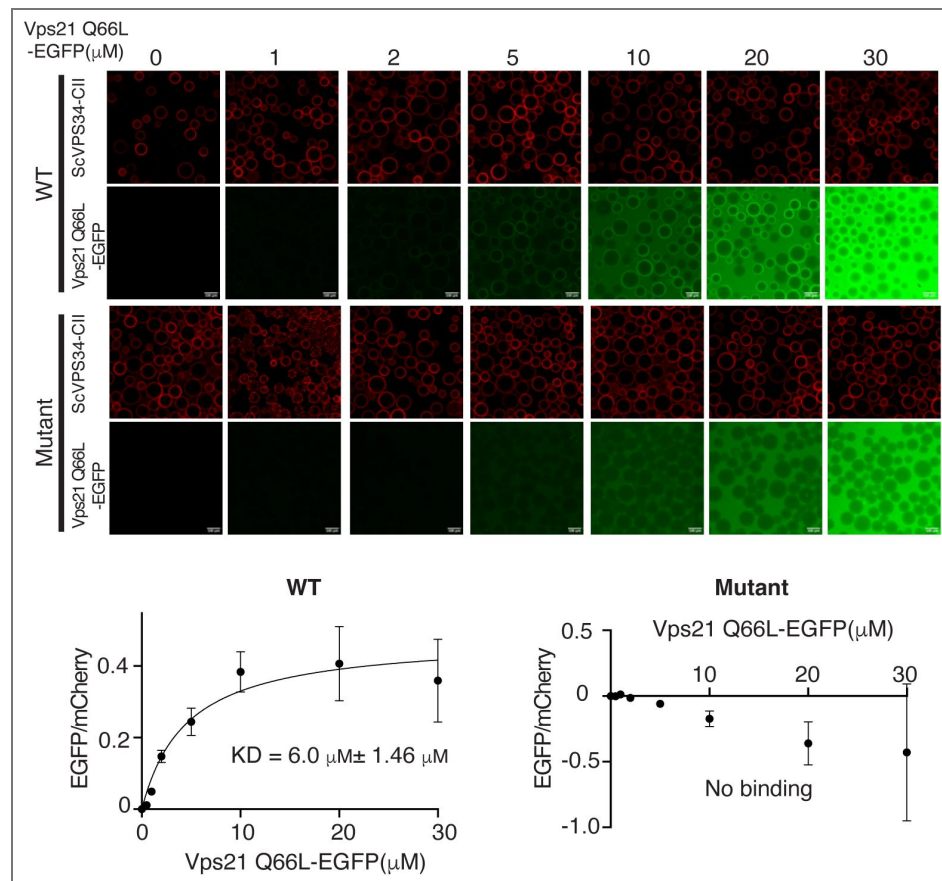


Figure S5. The RAB5 binding mutation in Vps15 in yeast VPS34-CII abolishes binding to Vps21, a yeast RAB5 orthologue.

A. Micrographs of RFP-trap bead-based binding assays for Vps21 Q66L-EGFP binding to mCherry-tagged yeast VPS34-CII (ScVPS34-CII). Top two rows: Vps21 Q66L-EGFP recruitment to WT ScVPS34-CII. Bottom two rows: Vps21 Q66L-EGFP recruitment to ScVPS34-CII carrying RAB5 binding mutant (Vps15 622-SHLITY-627 > DDLIEY). Micrographs are representatives from three independent experiments. Scale bars: 100 μm. **B.** Quantification of A. Representative quantifications from three independent experiments are shown. KD: Average ± SD from three independent experiments.

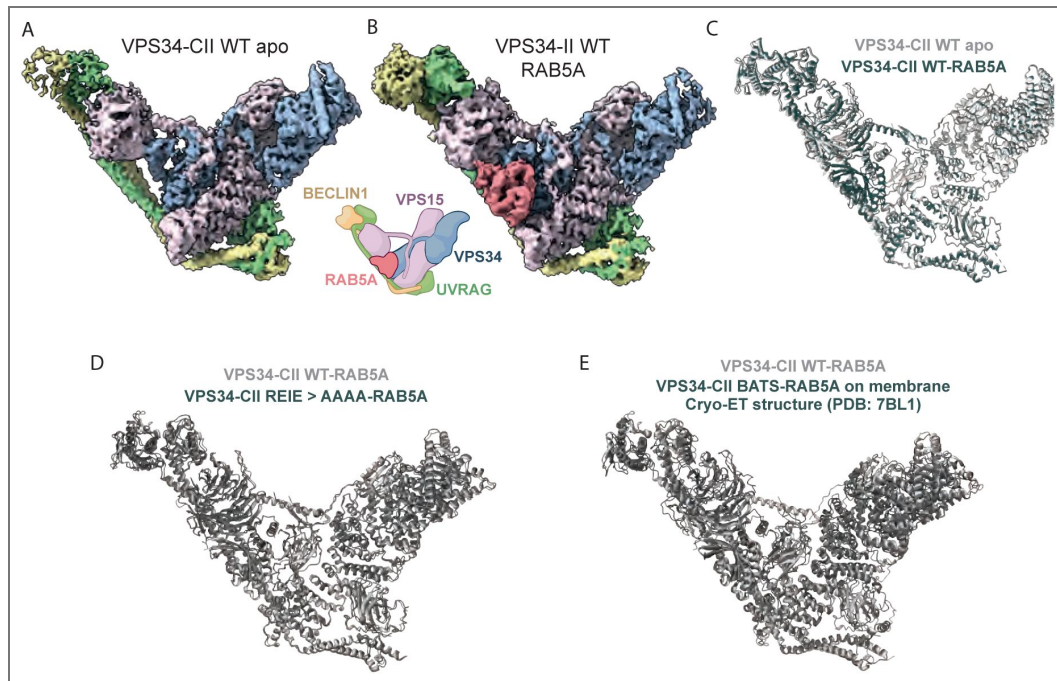


Figure S6. Human VPS34-CII with and without RAB5A has minimal conformational differences compared to other VPS34-CII structures.

Icons of VPS34-CII are based on cartoons from *BioRender*. **A.** Cryo-EM reconstruction of WT VPS34-CII apo form at 3.9 Å resolution. Particles contributing to this map do not have RAB5A bound and are a subset of all particles in the VPS34-CII + RAB5A sample (see materials and methods and Figure S11C). **B.** Cryo-EM reconstruction of WT VPS34-CII-RAB5A at 4.0 Å resolution. RAB5A is bound to VPS34 at the same site as in VPS34-CII carrying a VPS34 199-REIE-202>AAAA mutation (see Figure S2A). **C.** Overlay of atomic models of WT VPS34-CII apo (light grey), and WT VPS34-CII-RAB5A (dark grey). Models are aligned on the VPS15 helical solenoid (VPS15 residues 315-802), with RMSD of 3.3 Å across whole models. The conformation of the complex does not change substantially upon RAB5A binding. **D.** Overlay of the atomic models of WT VPS34-CII-RAB5A (light grey) and VPS34-CII REIE>AAAA-RAB5A (dark grey). Models are aligned on the VPS15 helical solenoid domain (VPS15 residues 315-802), with RMSD of 2.5 Å across whole models. **E.** Overlay of the atomic models of WT VPS34-CII-RAB5A (light grey), and cryo-ET derived model of VPS34-CII-BATS-RAB5A on the membrane (dark grey). Models were aligned on VPS34 kinase domain (VPS34 residues 539-887), and the backbone RMSD between VPS34, BECLIN1, UVRAG, and VPS15 WD40 domains from the two complexes is 7.0 Å. RMSD for VPS15 pseudokinase, helical solenoid, and SGD domains could not be measured accurately due to corrected VPS15 sequence register in the single particle EM-derived model of WT VPS34-CII-RAB5A.

Figure S7. Analysis of the cryo-EM structure of VPS34-CI-RAB1A (PDB 9MHG).

Icons of VPS34-CI are based on cartoons from *BioRender*. **A.** Close-up of RAB1A hydrophobic triad (F48 from Switch 1, W65 from Interswitch, and Y80 from Switch 2) with associated cryo-EM density. RAB1A residue W65 does not interact with VPS34-CI and instead stabilises the structure of RAB1A hydrophobic triad via interactions with RAB1A residues Q63 (from Interswitch region) and K13 (from outside the Switch regions). **B.** VPS34 residue R183 potentially makes hydrogen bonds with RAB1A residue S78 and the backbone carbonyls from RAB1A A68 and F73. Distances between hydrogen bond donor and acceptor atoms were measured with UCSF ChimeraX and fall within the range of distances of 2.2-4.0 Å compatible with hydrogen bonding (Jeirey, 1997).

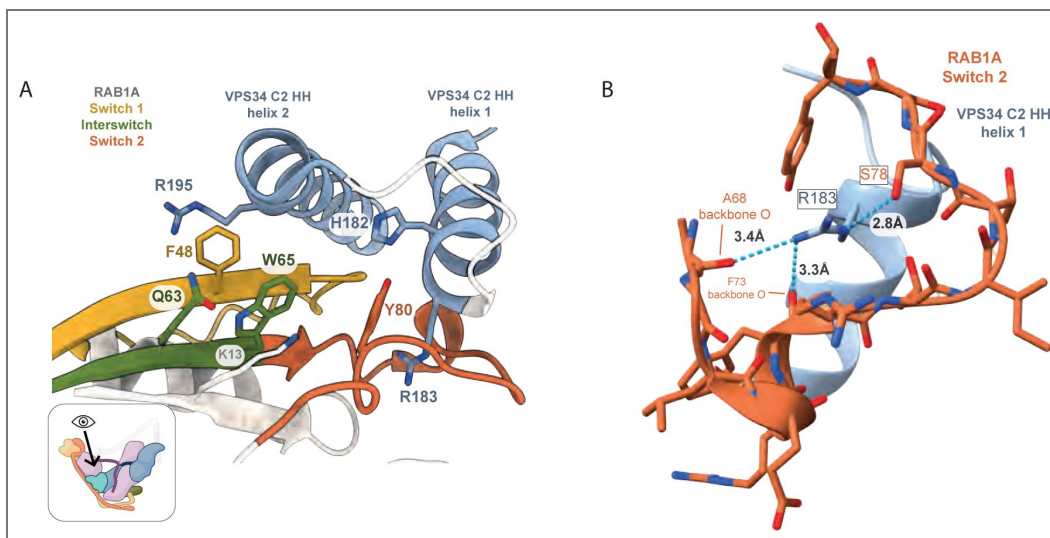


Figure S8. Multiple sequence alignment of selected human RAB GTPases and their isoforms.

Top two rows show RAB5A and RAB1A sequences for ease of comparison. Switch 1 and Switch 2 regions are highlighted with light blue background. Hydrophobic triad residues are indicated with arrows and framed in matching colour backgrounds. In Switch 1, there is an A > D change between RAB5A and RAB1A. In Switch 2, there is a P > S change between RAB5A and RAB1A. Residues in red background and white letters are identical; residues in red letters have conserved properties. The secondary structure elements depicted above the MSA are based on the crystal structure of RAB5A (PDB: 1N6H).

			$\alpha 1$	T.T	$\beta 2$	$\beta 3$	$\eta 1$	$\alpha 2$	$\beta 4$
			...				...		
				Switch 1: A > D	Triad: F	Triad: W	Switch 2: P > S	Triad: Y	
P20339	RAB5A_HUMAN/1-215	40	FVKGQFHE.FQES	FICAA	LTQSVCLDDTTVKFE	IWD	AG	ERYHSLAF	MYRCAQAAIVVY
P62820	RAB1A_HUMAN/1-205	31	FADDTYTE.SYIS	HIGVD	KIRTIELDGKTIKLO	IWD	AG	ERFRITTS	MYRCAHGIIVVY
P61020	RAB5B_HUMAN/1-215	40	FVKGQFHE.YQES	FICAA	LTQSVCLDDTTVKFE	IWD	AG	ERYHSLAF	MYRCAQAAIVVY
P51148	RAB5C_HUMAN/1-216	41	FVKGQFHE.YQES	FICAA	LTQSVCLDDTTVKFE	IWD	AG	ERYHSLAF	MYRCAQAAIVVY
Q9H0U4	RAB1B_HUMAN/1-201	28	FADDTYTE.SYIS	HIGVD	KIRTIELDGKTIKLO	IWD	AG	ERFRITTS	MYRCAHGIIVVY
Q92928	RAB1C_HUMAN/1-201	28	FADDPYTE.SYIS	HIGVD	KIRTIELDGKTIKLO	IWD	AG	ERFRITTS	MYRCAHGIIVVY
P51149	RAB7A_HUMAN/1-207	28	YVNNKFSN.QYKA	HIGAD	LTKEVMVDDRLVTMO	IWD	AG	ERFQSLGVAF	YKADCCVLVP
Q96AH8	RAB7B_HUMAN/1-199	28	YVHKTFYE.EYQT	HIGAD	ELSKIIILGDTTLKLO	IWD	AG	ERFRSMVSTF	YKSDGCIAP
P61006	RAB8A_HUMAN/1-207	28	FSEDAFNS.TFIS	HIGID	KIRTIELDGKKIKLO	IWD	AG	ERFRITTS	MYRCAHGIIVVY
Q92930	RAB8B_HUMAN/1-207	28	FSEDAFNT.TFIS	HIGID	KIRTIELDGKKIKLO	IWD	AG	ERFRITTS	MYRCAHGIIVVY
Q91026	RAB10_HUMAN/1-200	29	FSDDAFNT.TFIS	HIGID	KIKTVELQGGKIKLO	IWD	AG	ERFRITTS	MYRCAHGIIVVY
P62491	RAB11A_HUMAN/1-216	31	FTRNEFNL.ESKS	HIGVE	ATRSIQVVGKTIKLO	IWD	AG	ERFRITTS	MYRCAHGIIVVY
P61106	RAB14_HUMAN/1-215	31	FTEKKFMA.DCPH	HIGVE	ATRSIQVVGKTIKLO	IWD	AG	ERFRITTS	MYRCAHGIIVVY
Q9NP72	RAB18_HUMAN/1-206	28	FTEKTFDP.ELAA	HIGVE	ATRSIQVVGKTIKLO	IWD	AG	ERFRITTS	MYRCAHGIIVVY
Q969Q5	RAB24_HUMAN/1-203	27	YVHDFLVGPYQN	HIGAA	VAKVMSVGDRTVTLG	IWD	AG	ERYEAMSR	MYRCAKAAIVCY

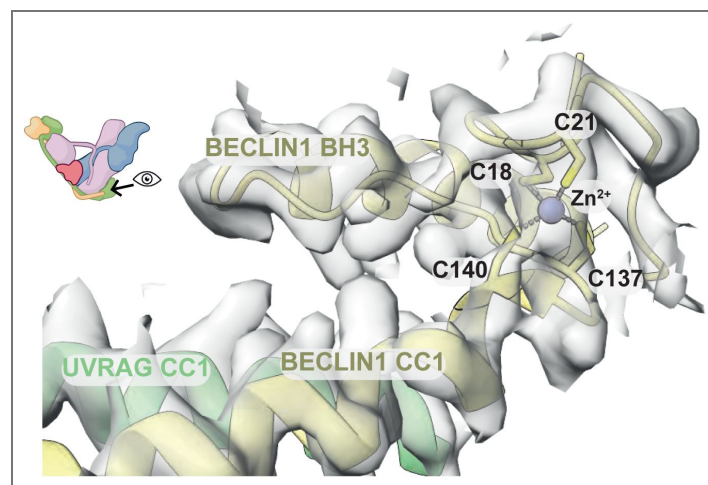


Figure S9. VPS34-CII has a zinc finger at BECLIN1 N-terminus.

Cryo-EM density of WT VPS34-CII-RAB5A at the BECLIN1 N-terminus. The zinc ion is coordinated by BECLIN1 residues C18 and C21, and C137 and C140.

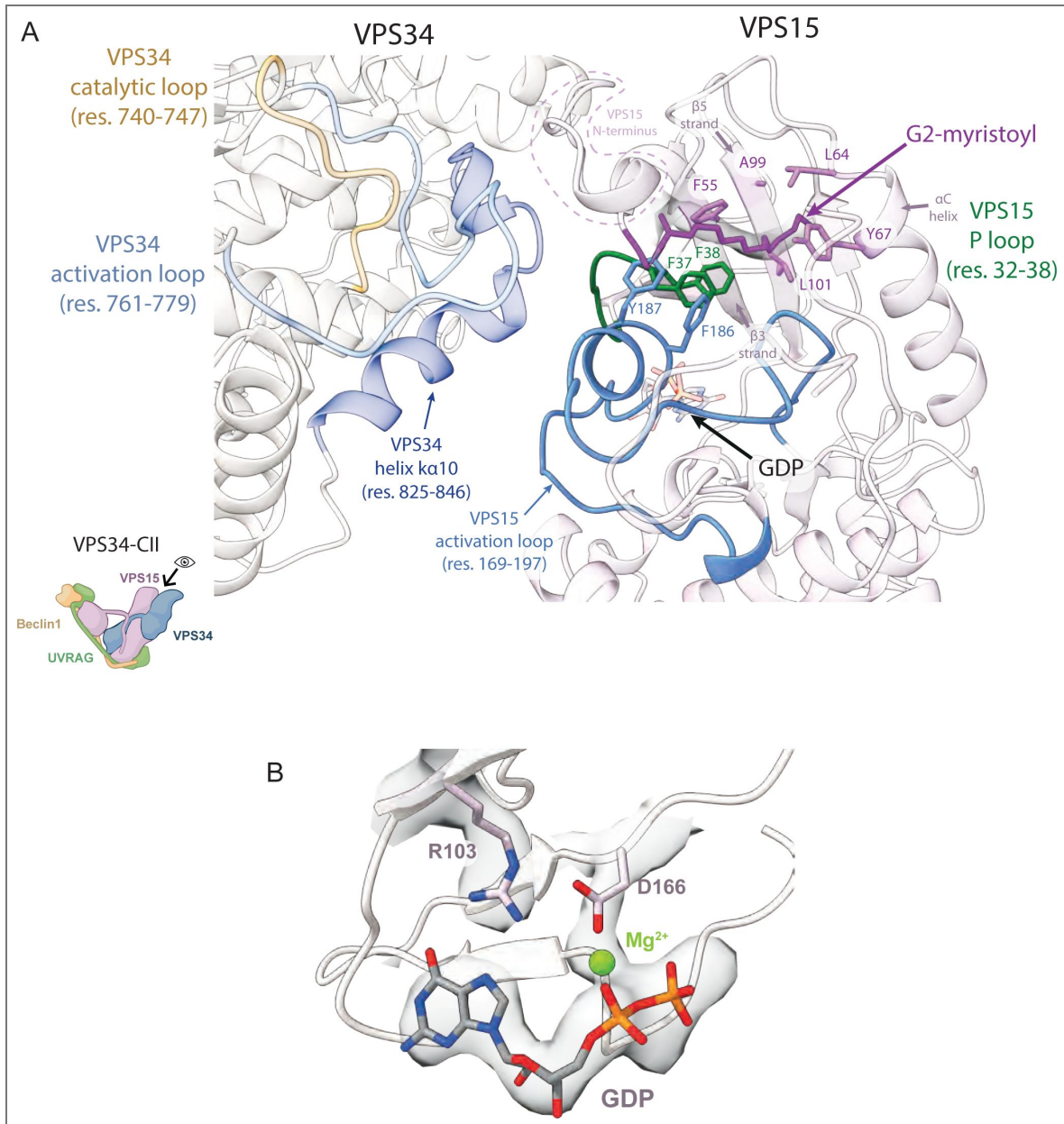


Figure S10. Cryo-EM densities show that the N-terminus of VPS15 is myristoylated and VPS15 is bound to a GDP nucleotide.

Icon of VPS34 complex is based on a cartoon from *BioRender*. **A.** VPS15 residue G2 is myristoylated (magenta). The myristoyl is inserted into a pocket formed of VPS15 N-lobe residues F55, A99, L101, L64, Y67 (purple), as well as P-loop residues F37 and F38 (green), and activation loop residues F186 and Y187 (blue). GDP nucleotide is close to the myristoyl pocket. VPS15 activation and P-loops are close to VPS34 helix α 10 (dark blue), but there is no contact with VPS34 catalytic loop (orange) or activation loop (light blue). The model is based on the high-resolution structure of VPS34-CII REIE>AAAA-RAB5A. **B.** Extra density in the VPS15 pseudokinase domain (VPS15 residues 24-315) is consistent with presence of a GDP nucleotide and a Mg^{2+} ion. VPS15 R103 contacts the GDP, and D166 coordinates the Mg^{2+} ion.

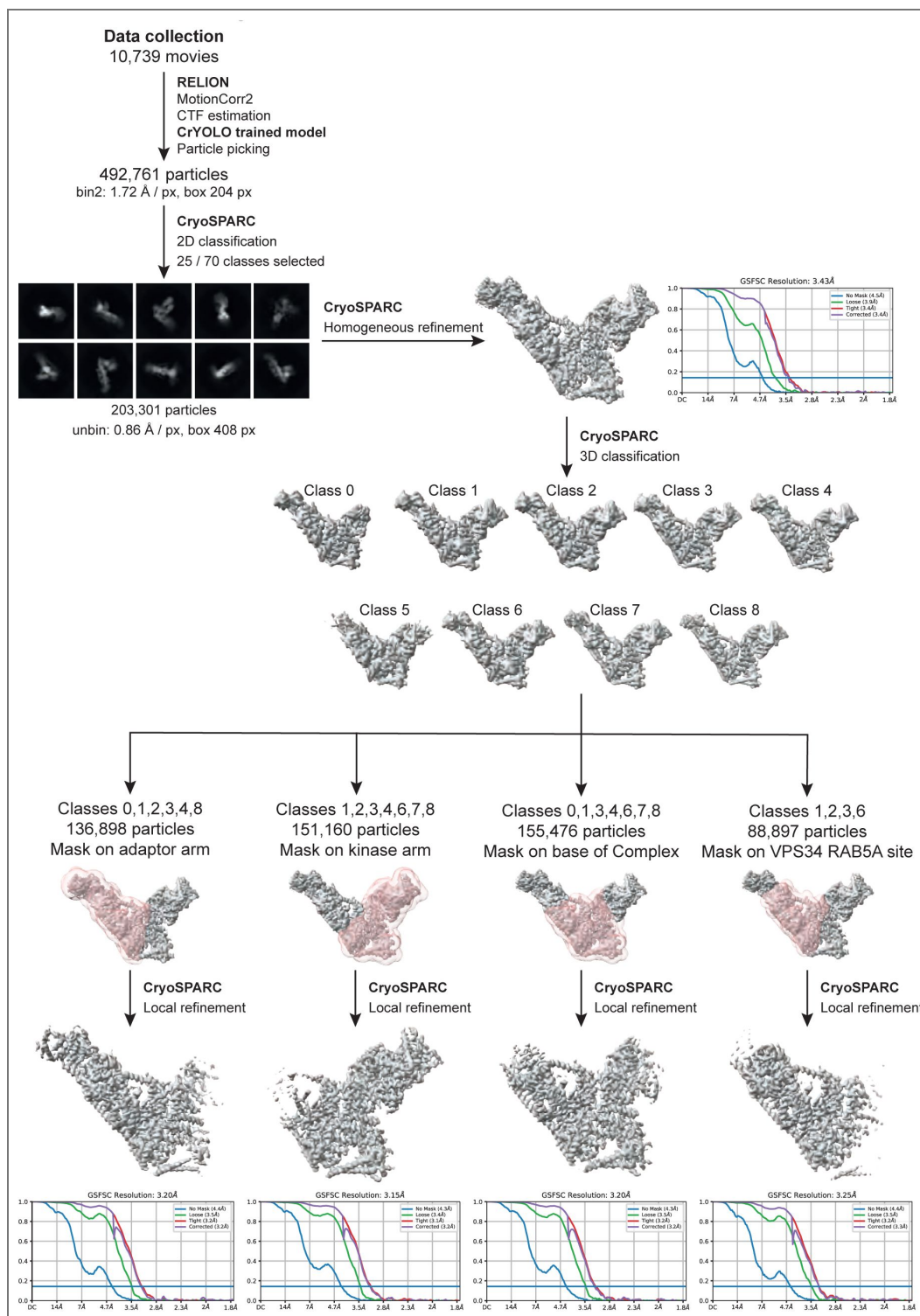


Figure S11A. Single particle cryo-EM data processing workflow for VPS34-CII REIE>AAAA-RAB5A (the 199-REIE-202 > AAAA mutation is in the VPS34 subunit).

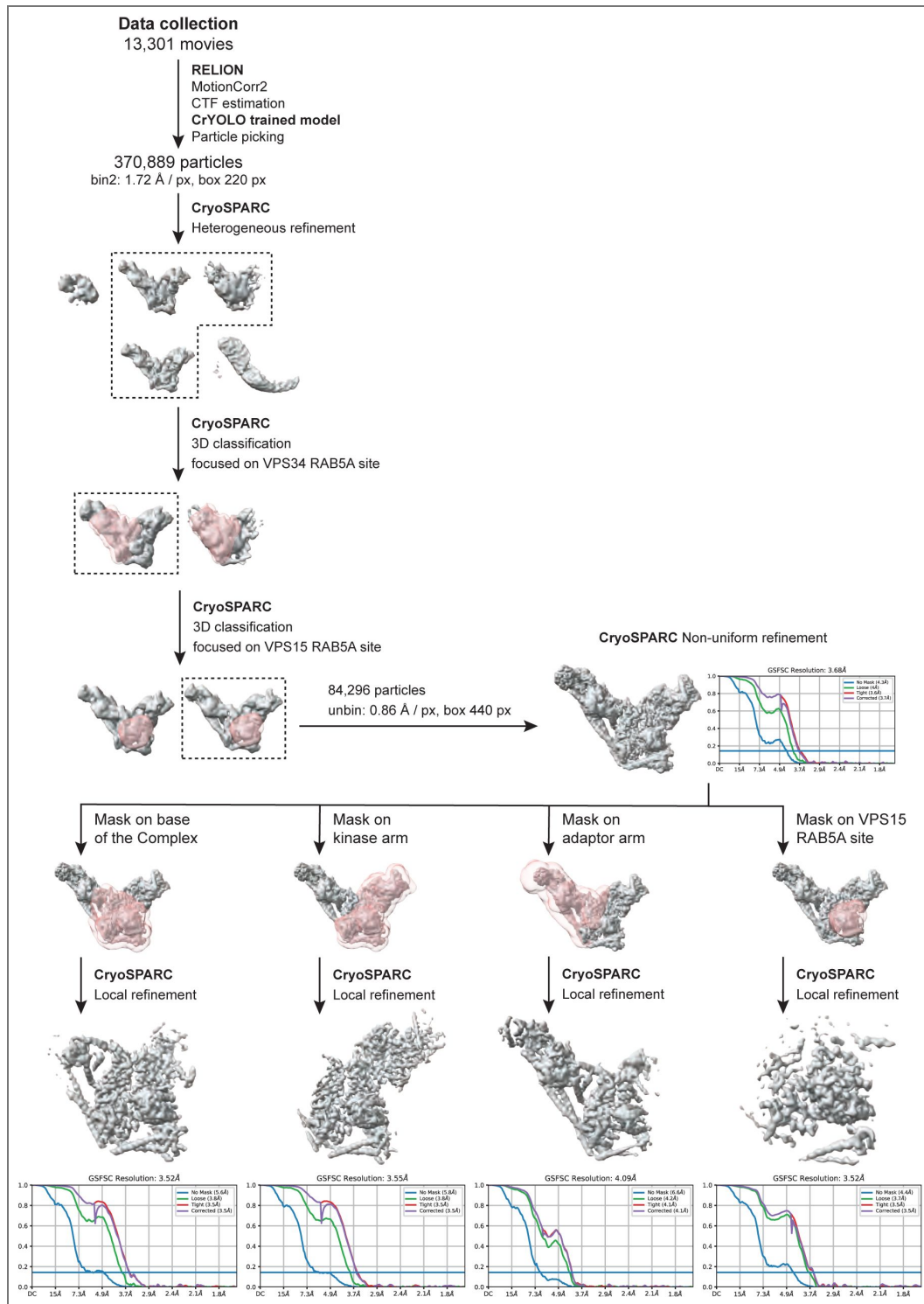


Figure S11B. Single particle cryo-EM data processing workflow for VPS34-CII REIE>ERIR-RAB5A (the 199-REIE-202 > ERIR mutation is in the VPS34 subunit).

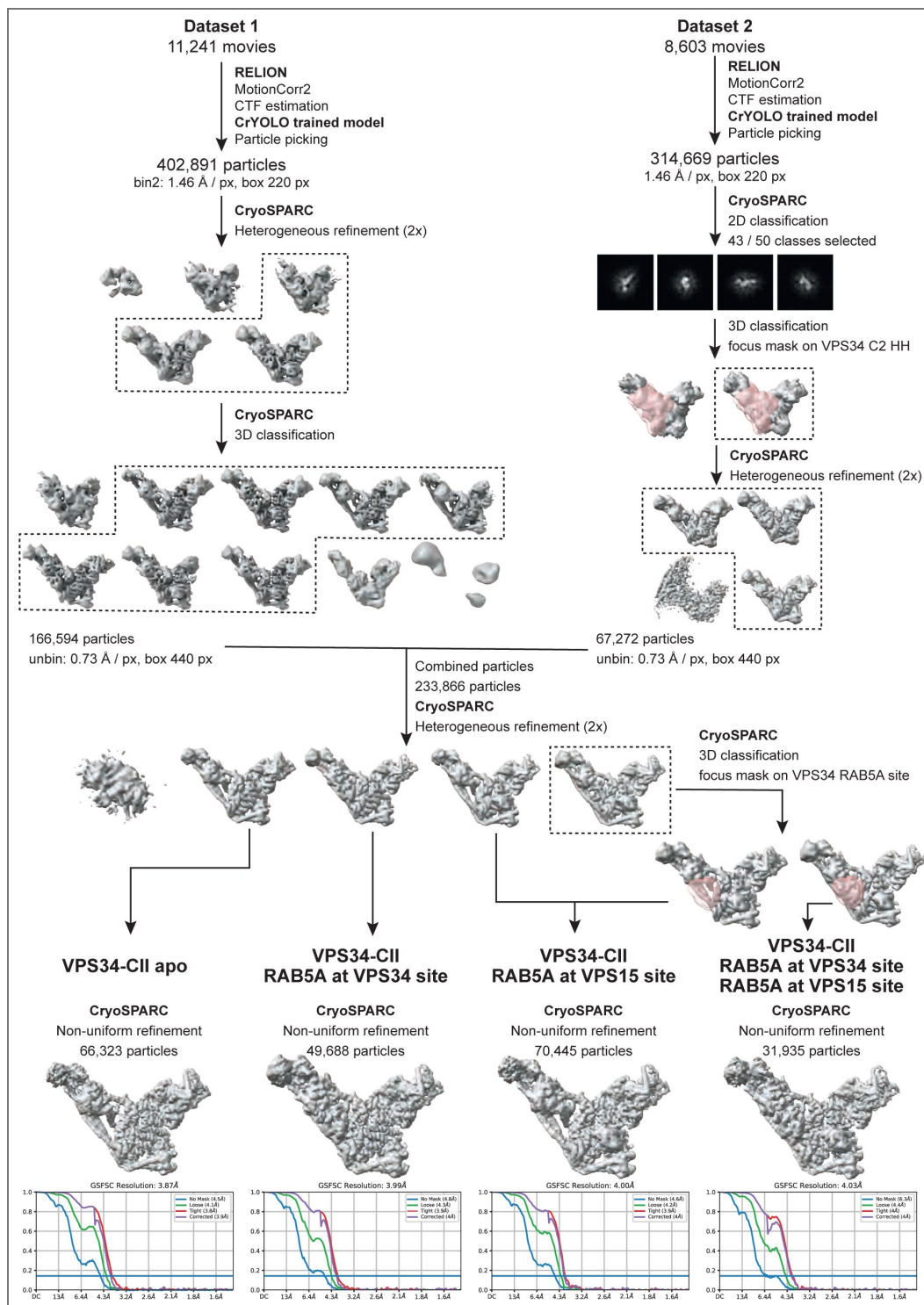


Figure S11C. Single particle cryo-EM data processing workflow for WT VPS34-CII - RAB5A.

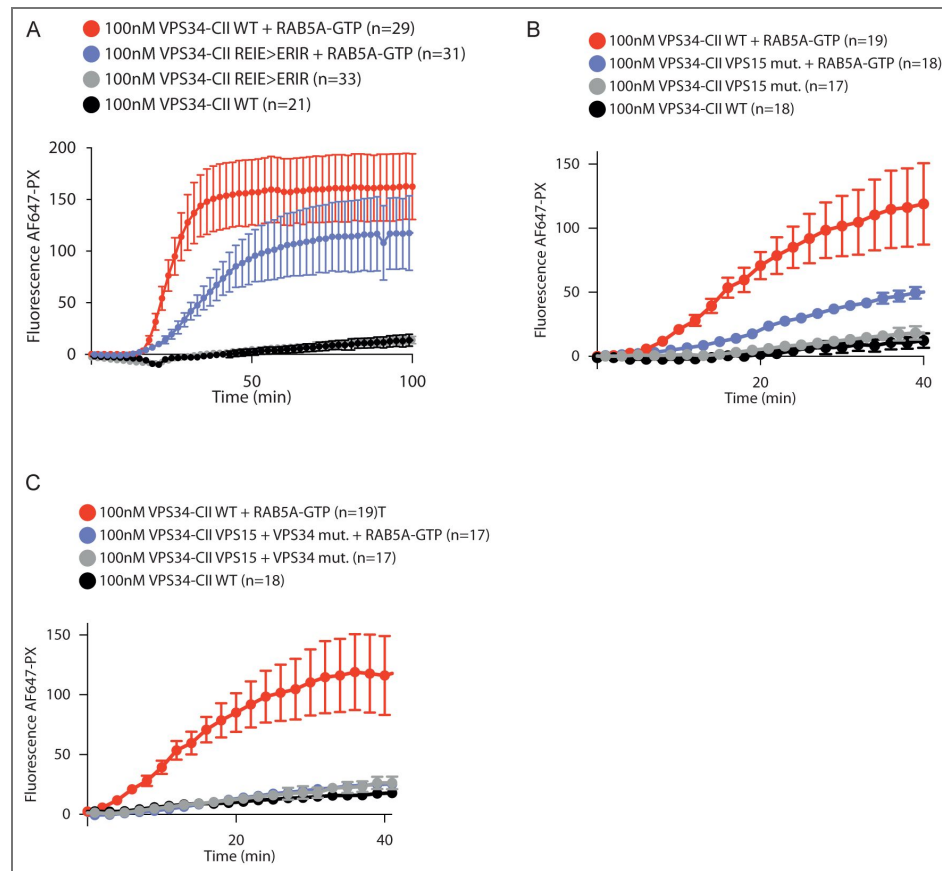


Figure S12. GUv-based kinase assay reaction curves with error bars for results shown in main figures.

'VPS34 mut.' denotes the 199-REIE-202 > ERIR mutation in human VPS34, and 'VPS15 mut.' denotes the 579-SHMIT-583 > DDMIE mutation in human VPS15. **A.** Reaction curves for the assay shown in Fig. 1C. **B.** Reaction curves for the assay shown in Fig. 2A. **C.** Reaction curves for the initial rate quantification shown in Fig. 2C.

apo complex II, with no density for RAB5A visible on VPS34 or VPS15 at any viewing threshold level on UCSF ChimeraX. These particles were used in non-uniform refinement and yielded a 3.87 Å reconstruction of apo complex II. 49,688 particles were assigned to the class with RAB5A bound to the VPS34 subunit. To confirm that in this particle set A) there were no particles with RAB5A bound to VPS15 and B) all particles had RAB5A bound to the VPS34 site, several rounds of ‘hard’ 3D classifications were performed with two classes and focus masks on the RAB5A binding site either on VPS34 or VPS15 subunits. No extra density on VPS15 was present, and none of the particles had an empty VPS34 RAB5A binding site; therefore, all particles were used in a non-uniform refinement job that yielded a 3.99 Å map.

44,127 particles contained complex II with the extra RAB5A density on the VPS15 subunit. These particles were further inspected using the procedure described above to ensure that A) no particles had RAB5A bound to the VPS34 subunit, and B) all particles had RAB5A bound to VPS15 subunit. This particle set was merged with the 26,318 particles found during the processing of the class with complex II + (RAB5A)₂ described below. The total of 70,445 particles for the density of complex II + RAB5A on the VPS15 subunit were used in non-uniform refinement that yielded a reconstruction at 4.00 Å, 58,253 particles that were classed as complex II with two RAB5A GTPases were used in a ‘hard’ 3D classification into two classes with a focus mask on the VPS34 RAB5A binding site. The class that did not have a RAB5A density on the VPS34 subunit contained 26,318 particles that were merged with the other 44,127 particles for complex II + RAB5A on VPS15 density. The other class with 31,935 particles was subject to further inspection as described above to confirm that all particles in this set A) had RAB5A bound to VPS15 and B) had RAB5A bound to the VPS34 site. All particles were confirmed to have two RAB5A proteins bound to complex II and were used for a non-uniform refinement job that produced a map at 4.03 Å resolution.

Maps from the consensus refinements were processed using CryoTEN. Models for complex II apo, complex II + RAB5A on VPS34, complex II + RAB5A on VPS15, and complex II + RAB5A on VPS34 and VPS15 were built by fitting a model of complex II REIE>AAAA + RAB5A into the density and adjusting the RAB5A chain appropriately. Models were refined using ISOLDE, StarMap, TEMPy-ReFF, PHENIX, and Servalcat. Models were validated through MolProbity. Consistent with what was reported previously for VPS34-CI(Cook *et al.*, 2025 [DOI](#)), our highest resolution cryo-EM structures contain a density continuous with the VPS15 N-terminus that is consistent with a myristate moiety. When this density was not apparent, the reconstructions had lower resolution. We also observed density for a GDP nucleotide in the VPS15 kinase domain. As it has been reported that VPS15 binds a GTP nucleotide in its active site(Cook *et al.*, 2025 [DOI](#)), it may be that the GTP had been hydrolysed, and we were seeing the bound product of the reaction (Fig. S10B [DOI](#)).

PDB models

Cryo-EM maps and refined VPS34-CII models were deposited in EMDB and PDB under accession codes:

WT VPS34-CII: PDB 9RX8, EMD-54361;

WT VPS34-CII-RAB5A_{VPS34}: PDB 9RX9, EMD-54362;

WT VPS34-CII-RAB5A_{VPS15}: PDB 9RXA, EMD-54363;

WT VPS34-CII-RAB5A_{VPS34,VPS15}: PDB 9RXB, EMD-54364;

VPS34-CII REIE>AAAA-RAB5A_{VPS34,VPS15}: PDB 9RX5, EMD-54358, EMD-54317, EMD-54316, EMD-54313, EMD-54312, EMD-54314, EMD-54315;

VPS34-CII REIE>ERIR-RAB5A_{VPS15}: PDB 9RX6, EMD-54359, EMD-54322, EMD-54321, EMD-54318, EMD-54319, EMD-54320;

Reprocessed tomography map for VPS34-CII(ΔBATS)-RAB5A_{VPS34} bound to membrane (Tremel *et al.*, 2021 [DOI](#)): EMD-54365

Data availability

Cryo-EM maps and refined VPS34-CII models were deposited in EMDB and PDB under accession codes: WT VPS34-CII: PDB 9RX8, EMD-54361; WT VPS34-CII-RAB5AVPS34: PDB 9RX9, EMD-54362; WT VPS34-CII-RAB5AVPS15: PDB 9RXA, EMD-54363; WT VPS34-CII-RAB5AVPS34,VPS15: PDB 9RXB, EMD-54364; VPS34-CII REIE>AAAA-RAB5AVPS34,VPS15: PDB 9RX5, EMD-54358, EMD-54317, EMD-54316, EMD-54313, EMD-54312, EMD-54314, EMD-54315; VPS34-CII REIE>ERIR-RAB5AVPS15: PDB 9RX6, EMD-54359, EMD-54322, EMD-54321, EMD-54318, EMD-54319, EMD-54320; Reprocessed tomography map for VPS34-CII(ΔBATS)-RAB5AVPS34 bound to membrane (Tremel et al., 2021 [DOI](#)): EMD-54365

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Additional files

[Summary of all peptides used for HDX-MS analysis](#) [DOI](#)

Additional information

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<https://doi.org/10.1038/nmeth.4193> | PubMed

Peer reviews

Reviewer #1 (Public review):

Summary:

This manuscript presents high resolution cryoEM structures of VPS34-complex II bound to Rab5A at 3.2Å resolution. The Williams group previously reported the structure of VPS34 complex II bound to Rab5A on liposomes using tomography, and therefore the previous structure, although very informative, was at lower resolution.

The first new structure they present is of the 'REIE>AAAA' mutant complex bound to RAB5A. The structure resembles the previously determined one except an additional molecule of RAB5A was observed bound to the complex in a new position, interacting with the solenoid of VPS15.

Although this second binding site exhibited reduced occupancy of RAB5A in the structure, the authors determined an additional structure in which the primary binding site was mutated to prevent RAB5A binding ('REIE>ERIR'). In this structure, there is no RAB5A bound to the primary binding site on VPS34, but the RAB5A bound to VPS15 now has strong density. The authors note that the way in which RAB5A interacts with each site is distinct, though both interfaces involve the switch regions. The authors confirm the location of this additional binding site using HDX-MS.

The authors then determine multiple structures of the wild-type complex bound to RAB5A from a single sample, as they use 3D classifications to separate out versions of the complex bound to 0, 1, or 2 copies of RAB5A. Overall the structure of VPS34-Complex II does not change between the different states, and the data indicate that both RAB5A binding sites can be occupied at the same time.

The authors then design a new mutant form of the complex (SHMIT>DDMIE) that is expected to disrupt the interaction at the secondary site between VPS15 and RAB5A. This mutation had a minor impact on the K_d for RAB5A binding, but when combined with the REIE>ERIR mutation of the primary binding site, RAB5A binding to the complex was abolished.

Comparison of sequences across species indicated that the RAB5A binding site on VPS15 was conserved in yeast while the RAB5A binding site on VPS34 is not.

The authors tested the impact of a correspond yeast Vps15 mutation (SHLITY>DDLIEY) predicted to disrupt interaction with yeast Rab5/Vps21, and found this mutant Vps15 protein was mislocalized and caused defective CPY processing.

The authors then compare these structures of the RAB5A-class II complex to recently published structures from the Hurley group of the RAB1A-class I complex, and find that in both complexes the Rab protein is bound to the VPS34 binding site in a somewhat similar manner. However, a key difference is the position of VPS34 is slightly different in the two complexes because of the unique ATL14L and UVRAG subunits in the class I and class II

complexes, respectively. This difference creates a different RAB binding pocket that explains the difference in RAB specificity between the two complexes.

Finally, the higher resolution structures enable the authors to now model portions of BECLIN1 and UVRAG that were not previously modeled in the cryoET structure.

Strengths:

Overall I found this to be an interesting and comprehensive study of the structural basis for interaction of RAB5A with VPS34-complex II. The authors have performed experiments to validate their structural interpretations, and they present a clear and thorough comparative analysis of the Rab binding sites in the two different VPS34 complexes. The result is a much better understanding of how two different Rab GTPases specifically recruit two different, but highly similar complexes to the membrane surface.

Weaknesses:

No significant weaknesses noted.

<https://doi.org/10.7554/eLife.110040.2.sa3>

Reviewer #2 (Public review):

The work by Spokaite et al describes the discovery of a novel Rab5 binding site present in complex II of class III PI3K using a combination of HDX and Cryo EM. Extensive mutational and sequence analysis define this as the primordial Rab5 interface. The data presented are convincing that this is indeed a biologically relevant interface, and is important in defining mechanistically how vps34 complexes are regulated.

This paper is a very nice expansion of their previous cryo-ET work from 2021, and is an excellent companion piece on high resolution cryo-EM of the complex I class III complex bound to Rab1 from the Hurley lab in 2025. Overall, this work is of excellent technical quality, and answers important unexplained observations on some unexpected mutational analysis from the previous work.

They used their increased affinity vps34 mutant to determine the 3.2 ang structure of Rab5 bound to vps34-CII. Clear density was seen for the original Rab5 interface, but an additional site was observed. Based on this structure they mutated out the vps34 interface, allowing for a high resolution structure of the Rab5 bound at the Vps15 interface.

They extensively validated the vps15 interface in the yeast variant of vps34, showing that the Vp215-Rab5 (Vps21) interface identified is critical in controlling complex II vps34 recruitment.

The major strengths of this paper are that the experiments appear to be done carefully and rigorously and I have very few experimental suggestions.

Here is what I recommend based on some very minor weaknesses I observed

(1) My main concern has to do a little bit with presentation. My main issue is how the authors use mutant description. They clearly indicate the mutant sequence in the human isoform (for example see Fig 2A, Vps15 described as 579-SHMIT-583>DDMIE), however, when they shift to the yeast version they shift to saying vps15 mutant, but don't define the mutant, Fig 2G). I would recommend they just include the same sequence numbering and WT to mutant replacement every time a new mutant (or species) is described. It is always easier to interpret what is being shown when the authors are jumping between species when the exact mutant is included. This is particularly important in this paper, where we are jumping between both

different subunits and different species, so clear description in figure/figure legends makes it much easier to read for non-specialists.

(2) The HDX data very clearly shows that Rab5 is likely able to bind at both sites, which back ups the cryo EM data nicely. I am slightly confused by some of the HDX statements described in the methods.

(3) The authors state "Only statistically significant peptides showing a difference greater than 0.25 Da and greater than 5% for at least two timepoints were kept." This seems to be confusing why they required multiple timepoints, and before they also describe that they required a p value of less than 0.05. It might be clearer to state that significant differences required a 0.25 Da, 5%, and p value of <0.05 (n=3). Also what do they mean by kept? Does this mean that they only fully processed the peptides with differences.

(4) They show peptide traces for a selection in the supplement, but it would be ideal to include the full set of HDX data as an excel file, including peptides with no differences as there is a lot of additional information (deuteration levels for everything) that would be useful to share, as recommended from the Masson et al 2019 recommendations paper. This may be attached but this reviewer could not see an example of it in the shared data dropbox folder.

Comments on revisions:

The authors have addressed all of my issues.

<https://doi.org/10.7554/eLife.110040.2.sa2>

Reviewer #3 (Public review):

Summary:

The manuscript of Spokaite et al. focuses on the Vps34 complex involved in PI3P production. This complex exists in two variants, one (class I) specific for autophagy, and a second one (class II) specific for the endocytic system. Both differ only in one subunit. The authors previously showed that the Vps34 complexes interact with Rab GTPases, Rab1 or Rab5 (for class II), and the identified site was found at Vps34. Now, the authors identify a conserved and overlooked Rab5 binding site in Vps15, which is required for the function of the Class II complex. In support of this, they show cryo-EM data with a second Rab5 bound to Vps15, identify the corresponding residues, and show by mutant analysis that impaired Rab5 binding also results in defects using yeast as a model system.

Overall, this is a most complete study with little to criticize. The paper shows convincingly that the two Rab5 binding sites are required for Vps34 complex II function, with the Vps15 binding site being critical for endosomal localization. The structural data is very much complete. What I am missing are a few controls that show that the mutations in Vps15 do not affect autophagy. I also found the last paragraph of the results section a bit out of place, even though this is a nice observation that the N-terminal part of BECLIN has these domains. However, what does it add to the story?

Comments on revisions:

The authors answered all my questions. I have no further requests.

<https://doi.org/10.7554/eLife.110040.2.sa1>

Author Response:

The following is the authors' response to the original reviews.

Public Reviews:

Reviewer #1 (Public review):

Summary:

This manuscript presents high-resolution cryoEM structures of VPS34-complex II bound to Rab5A at 3.2Å resolution. The Williams group previously reported the structure of VPS34 complex II bound to Rab5A on liposomes using tomography, and therefore, the previous structure, although very informative, was at lower resolution.

The first new structure they present is of the 'REIE>AAAA' mutant complex bound to RAB5A. The structure resembles the previously determined one, except that an additional molecule of RAB5A was observed bound to the complex in a new position, interacting with the solenoid of VPS15.

Although this second binding site exhibited reduced occupancy of RAB5A in the structure, the authors determined an additional structure in which the primary binding site was mutated to prevent RAB5A binding ('REIE>ERIR'). In this structure, there is no RAB5A bound to the primary binding site on VPS34, but the RAB5A bound to VPS15 now has strong density. The authors note that the way in which RAB5A interacts with each site is distinct, though both interfaces involve the switch regions. The authors confirm the location of this additional binding site using HDX-MS.

The authors then determine multiple structures of the wild-type complex bound to RAB5A from a single sample, as they use 3D classifications to separate out versions of the complex bound to 0, 1, or 2 copies of RAB5A. Overall, the structure of VPS34-Complex II does not change between the different states, and the data indicate that both RAB5A binding sites can be occupied at the same time.

The authors then design a new mutant form of the complex (SHMIT>DDMIE) that is expected to disrupt the interaction at the secondary site between VPS15 and RAB5A. This mutation had a minor impact on the K_d for RAB5A binding, but when combined with the REIE>ERIR mutation of the primary binding site, RAB5A binding to the complex was abolished.

Comparison of sequences across species indicated that the RAB5A binding site on VPS15 was conserved in yeast, while the RAB5A binding site on VPS34 is not.

The authors tested the impact of a corresponding yeast Vps15 mutation (SHLITY>DDLIEY) predicted to disrupt interaction with yeast Rab5/Vps21, and found that this mutant Vps15 protein was mislocalized and caused defective CPY processing.

The authors then compare these structures of the RAB5A-class II complex to recently published structures from the Hurley group of the RAB1A-class I complex, and find that in both complexes the Rab protein is bound to the VPS34 binding site in a somewhat similar manner. However, a key difference is that the position of VPS34 is slightly different in the two complexes because of the unique ATL14L and UVRAG subunits in the class I and class II complexes, respectively. This difference creates a different RAB binding pocket that explains the difference in RAB specificity between the two complexes.

Finally, the higher resolution structures enable the authors to now model portions of BECLIN1 and UVRAG that were not previously modeled in the cryoET structure.

Strengths:

Overall, I found this to be an interesting and comprehensive study of the structural basis for the interaction of RAB5A with VPS34-complex II. The authors have performed experiments to validate their structural interpretations, and they present a clear and thorough comparative analysis of the Rab binding sites in the two different VPS34 complexes. The result is a much better understanding of how two different Rab GTPases specifically recruit two different, but highly similar complexes to the membrane surface.

Weaknesses:

No significant weaknesses were noted.

Reviewer #2 (Public review):

Summary:

The work by Spokaite et al describes the discovery of a novel Rab5 binding site present in complex II of class III PI3K using a combination of HDX and Cryo EM. Extensive mutational and sequence analysis define this as the primordial Rab5 interface. The data presented are convincing that this is indeed a biologically relevant interface, and is important in defining mechanistically how VPS34 complexes are regulated.

This paper is a very nice expansion of their previous cryo-ET work from 2021, and is an excellent companion piece on high-resolution cryo-EM of the complex I class III complex bound to Rab1 from the Hurley lab in 2025. Overall, this work is of excellent technical quality and answers important unexplained observations on some unexpected mutational analysis from the previous work.

They used their increased affinity VPS34 mutant to determine the 3.2 ang structure of Rab5 bound to VPS34-CII. Clear density was seen for the original Rab5 interface, but an additional site was observed. Based on this structure, they mutated out the VPS34 interface, allowing for a high-resolution structure of the Rab5 bound at the VPS15 interface.

They extensively validated the VPS15 interface in the yeast variant of VPS34, showing that the Vp215-Rab5 (VPS21) interface identified is critical in controlling complex II VPS34 recruitment.

The major strengths of this paper are that the experiments appear to be done carefully and rigorously, and I have very few experimental suggestions.

Here is what I recommend based on some very minor weaknesses I observed

(1) My main concern has to do a little bit with presentation. My main issue is how the authors use mutant description. They clearly indicate the mutant sequence in the human isoform (for example, see Figure 2A, VPS15 described as 579-SHMIT-583>DDMIE); however, when they shift to the yeast version, they shift to saying VPS15 mutant, but don't define the mutant, Figure 2G). I would recommend they just include the same sequence numbering and WT to mutant replacement every time a new mutant (or species) is described. It is always easier to interpret what is being shown when the authors are jumping between species, when the exact mutant is included. This is particularly important in this paper, where we are jumping between different subunits and different species, so a clear description in the figure/figure legends makes it much easier to read for non-specialists.

The reviewer has made an excellent point here. To clarify the yeast mutation, we have revised the manuscript main text to refer to the yeast mutant as SHLITY>DDLIEY, and we

have added this to the legend for Figs. 2F,G.

(2) The HDX data very clearly shows that Rab5 is likely able to bind at both sites, which backs up the cryo EM data nicely. I am slightly confused by some of the HDX statements described in the methods.

(3) The authors state, "Only statistically significant peptides showing a difference greater than 0.25 Da and greater than 5% for at least two timepoints were kept." This seems to be confusing as to why they required multiple timepoints, and before they also describe that they required a p-value of less than 0.05. It might be clearer to state that significant differences required a 0.25 Da, 5%, and p-value of <0.05 ($n=3$). Also, what do they mean by kept? Does this mean that they only fully processed the peptides with differences?

(4) They show peptide traces for a selection in the supplement, but it would be ideal to include the full set of HDX data as an Excel file, including peptides with no differences, as there is a lot of additional information (deuteration levels for everything) that would be useful to share, as recommended from the Masson et al 2019 recommendations paper. This may be attached, but this reviewer could not see an example of it in the shared data dropbox folder.

We have revised the HDX method description to clarify. All peptides were kept and fully processed. However, for the results displayed, we have illustrated only peptides meeting the criteria described.

The Excel file for all peptides (as recommended by Masson et al) was deposited with PRIDE, with the identifier with the dataset identifier PXD061277, in addition, we have included this excel file in our supplementary material.

Reviewer #3 (Public review):

Summary:

The manuscript of Spokaite et al. focuses on the Vps34 complex involved in PI3P production. This complex exists in two variants, one (class I) specific for autophagy, and a second one (class II) specific for the endocytic system. Both differ only in one subunit. The authors previously showed that the Vps34 complexes interact with Rab GTPases, Rab1 or Rab5 (for class II), and the identified site was found at Vps34. Now, the authors identify a conserved and overlooked Rab5 binding site in Vps15, which is required for the function of the Class II complex. In support of this, they show cryo-EM data with a second Rab5 bound to Vps15, identify the corresponding residues, and show by mutant analysis that impaired Rab5 binding also results in defects using yeast as a model system.

Overall, this is a most complete study with little to criticize. The paper shows convincingly that the two Rab5 binding sites are required for Vps34 complex II function, with the Vps15 binding site being critical for endosomal localization. The structural data is very much complete.

Weaknesses:

What I am missing are a few controls that show that the mutations in Vps15 do not affect autophagy. I am wondering if this mutant is still functional in autophagy. This can be simply tested by sorting of Atg8 to the vacuole lumen using established assays or by following PhoΔ60 sorting. This analysis would reveal that the corresponding mutant is specific for the Class II complex.

One of the first noted features of the VPS34 complexes was that the ATG14-containing complex (VPS34-CI) is important for autophagy, while the VPS38 (yeast orthologue of UVRAG) subunit characteristic of VPS34-CII is important for endocytic sorting (PMID 11157979).

However, the VPS34, VPS15 and BECLIN1 subunits are required are present in both complexes, as such, mutations of them may affect both processes.

We agree with the reviewer that is an important undertaking to examine the effect of the SHLITY>DDLIEY mutation in yeast Vps15 on autophagy. However, the focus of the current manuscript is VPS34-complex II and RAB5 interaction/activation. An autophagy effect would be more relevant for VPS34 complex I and RAB1. We have not presented any results for human VPS34-complex I - RAB1 nor yeast Vps34-complex I – Ypt1 (yeast RAB1 orthologue). We are preparing another manuscript focusing entirely on this, and it is not a simple story. While we think this is an important question, we believe that this is beyond the scope of the current manuscript.

It would be helpful if the authors could clarify whether they believe that Vps34 kinase activity is stimulated by Rab binding or whether this stimulation is a consequence of better membrane localization of Vps34. In other words, is the complex active with soluble PI3P in solution, and does the activity change if Rab5 is added to the complex? This might have been addressed in the past, but I did not see evidence for this, as the authors only addressed the activity of the Vps34 complexes on membranes.

The reviewer has raised an excellent question, which was addressed briefly in the introduction to the manuscript. We have now somewhat expanded on these issues near the end of the discussion in the revised manuscript. In our previously published study, we found that soluble RAB5-GTP did not stimulate the complex II activity (supplementary figure 2b of PMID: 33692360). This is consistent with our finding in this manuscript showing that RAB5 did not cause large conformational changes in solution. However, our previous single-molecule study showed that once complex II is recruited to the membrane by RAB5, and RAB5 increases the turnover rate on membranes, indicating an additional allosteric activation (Figure 7 of PMID: 33137306). This study indicated that the primary role of RAB5 is to anchor complex II on the membrane. Once the complex is anchored on the membrane by RAB5, the kinase domain is in the vicinity of its substrate, PI, leading to higher turnover.

The Echelon Class III PI3K ELISA Kit (Echelon, K-3000) comes with a soluble PI, diC8 to measure the VPS34 activity, and it is certainly active with this soluble substrate. However, if the substrate is in membranes, the VPS34 activity is greatly dependent on the character of the membrane.

I also found the last paragraph of the results section a bit out of place, even though this is a nice observation that the N-terminal part of BECLIN has these domains. However, what does it add to the story?

The reviewer is correct that the high-resolution features of BECLIN1 at the base of the V-shaped complex that we observed are not related to RAB5 binding, but they are characteristic of VPS34-CII and likely to be important for the specific role of VPS34-CII. This is the first high-resolution structure of the VPS34-CII that has been reported, and we believe it would be irresponsible not to briefly describe them, since they are unique to VPS34-CII. For this reason, we have placed this section at the end of the results, and we now clarify that we do not see a relevance to RAB5 function, but we describe the arrangement of a region (the BH3) that has been functionally noted in many previous studies, in the absence of a structure.

Reviewing Editor Comments:

Please address the following suggestions for minor changes to the manuscript. Use your best scientific judgment in addressing the comments and describe the modifications together with your reasoning in a cover letter. We look forward to seeing the revised version of this very nice study.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

I found a portion of the description of the cryoEM complexes on the top of page 9 to be redundant with similar descriptions near the top of page 7, and it was not clear to me at first that these were describing the same structures. Part of my confusion was due to the redundancy, including the statement near the bottom of page 7: 'Models were built and refined for all RAB5-associated VPS34-CII assemblies', and then the similar statement on page 9: 'We fit and refined atomic models into both densities'. I believe these are describing the same models? To clarify for the reader, perhaps on page 9, the authors could begin this part with a statement such as "as described above", and eliminate the redundant descriptions.

The reviewer is correct. Both sections describe the same set of cryo-EM classes from the same sample. The only difference is what we analysed in the two sections: number of RAB5s bound in the first section and the effect of RAB5 binding in the second section. We have revised the text to make this clear, and to make the second section more succinct.

Reviewer #3 (Recommendations for the authors):

(1) The authors show nicely that a mutation in Vps15 disrupts binding to Vps21 in vivo, with defects in the endocytic pathway as analyzed by CPY sorting. I am wondering if this mutant is still functional in autophagy. This can be simply tested by sorting of Atg8 to the vacuole lumen using established assays or by following PhoΔ60 sorting. This analysis would reveal that the corresponding mutant is specific for the Class II complex. If the authors were to find evidence that this Vps15 mutant also affects autophagy, it would indicate that there is possibly also another Rab1 binding site in Vps15.

As we stated above, an autophagy effect would be more relevant for VPS34 complex I and RAB1. We have not presented any results for human VPS34-complex I - RAB1 nor yeast Vps34-complex I – Ypt1 (yeast RAB1 orthologue). We are preparing another manuscript focusing entirely on this, and it is not a simple story. While we think this is an important question, we believe that this is beyond the scope of the current manuscript.

(2) It would be helpful if the authors could clarify whether they believe that Vps34 kinase activity is stimulated by Rab binding or whether this stimulation is a consequence of better membrane localization of Vps34. In other words, is the complex active with soluble PI3P in solution, and does the activity change if Rab5 is added to the complex? This might have been addressed in the past, but I did not see evidence for this, as the authors only addressed the activity of the Vps34 complexes on membranes.

As in our response to reviewer #3 above, this point was addressed in previous publications and was described in the introduction to our manuscript.

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